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Excess Thermodynamic Properties of Solutions in *Ln-Ru* (*Ln* = *Nd*, *Gd*, *Dy*) Binary Systems Based on Quadratic, Exponential and Combined Models Supported by Ab-Initio Calculations

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Abstract The systematic thermodynamic optimization of the *Ln-Ru* (where *Ln* = *Nd*, *Gd*, and *Dy*) binary systems is performed using the *CALPHAD* approach. Using this approach, experimental information published in literature, including both phase equilibria and thermodynamic data, are critically evaluated. The thirteen intermediate phases with stoichiometries, such as Ln_3Ru, Nd_5Ru_2, Ln_7Ru_3, Nd_5Ru_3, $Ln_{73}Ru_{27}$, Ln_2Ru, and $LnRu_2$, which have been reported for the three binary *Ln-Ru* systems are considered as stoichiometric compounds. An exponential and a quadratic model are used to describe the temperature dependence of the excess quantities for the Liquid, (αNd), (βNd), (αDy), (βDy), (αGd), (βGd), and (Ru) solution phases. The results were compared with those from a combined quadratic-exponential temperature dependence description of the excess energies. Using the Thermo-Calc software, self-consistent sets of thermodynamic parameters were obtained to describe the Gibbs energies of the numerous phases in the *Ln-Ru* binary systems. The calculated results are in good agreement with the available phase equilibria and thermodynamic data.

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1 Introduction

The complexity of separating rare earth elements (*REEs*) into indigenous metals and earths due to their pronounced earthy appearance as oxides, resulted in attribution of this denomination.[1] Several environmental issues are related to their extraction because their ore minerals also contain *Ca*, *Th*, and *U*.[2] Recently, the European Commission has classified some *REEs* as critical materials with a significant risk to supply.[3] Contemporary research spotlights *Nd*, *Gd*, and *Dy* as the most critical of all of the *REEs* by virtue of their use in magnetic and phosphor materials.[4] Primary mining of *REE* ores for *neodymium* produces an excess of the plenteous elements, *lanthanum* and *cerium* and makes *REEs* recycling necessary. Therefore, recycling of *neodymium* can lessen the total amount of *REE* ores that ought to be extracted. Bonded together with transition elements, they exhibit an excellent combination of thermal and magnetic properties, and can be used in a wide field of applications using new materials, such as the materials for hydrogen storage,[5,6] permanent magnets,[7] lamp phosphors, rechargeable *NiMH* batteries, and catalysts.[8] Research is ongoing to understand the alloying behavior of rare earths, such as *cerium*, *neodymium*, *samarium*, *gadolinium*, and *dysprosium* with *ruthenium* in view of possible applications. Therefore, the phase relations provide important guidance for material design and are an essential basis for the *CALPHAD* approach.[9–14] In particular, knowledge of the phase diagrams and the

thermodynamic properties of the *Ln-Ru* systems is an essential task. The remedy to this problem is the study of the *Ln-Ru-X* ($X = Si$ or Ge) ternary systems, which are known to display a number of crucial properties such as heavy-fermion superconductivity, Kondo behavior, anomalous magnetism, and/or intermediate valency. Thus, it is imperative to begin with the assessment of their constituting binary systems. Our thermodynamic optimization of these binary systems is performed based on the experimental data available in the literature, including phase diagrams, thermodynamic measurements, and predicted data such as enthalpies of formation. In this case, it may be required to work with optimization programs that contain more advanced modelling, such as Thermo-Calc software refined by Jansson[15] and Sundman et al.[16] Moreover, Chen et al.[17] discussed the appearance of artificial inverted miscibility gaps in many calculated phase diagrams already published in literature. Theoretically, Kaptay[18] and Abe et al.[19] found that these behaviors arise if the linear model is applied, and if $a_0^{\Phi} < 0$, $b_0^{\Phi} > 2 \times R$. Kaptay,[20] based on the Lupis–Elliott rule, quite generally suggests that an increase in the temperature tends to bring a system closer to ideality. This statement is in accord with the heat of mixing and an excess entropy of mixing of the same solution should have the same signs. Kaptay,[18] suggested a new function to describe the excess Gibbs energy of the liquid and solution phases by assuming that all excess properties tend to be towards zero as the temperature approaches infinity. From another case, Schmid-Fetzer et al.[13] noted that the application of the Kaptay equation to model the temperature dependence of the interaction parameters in the liquid and solid solution phases results in artificial phase stabilities at low temperatures. Kaptay[18,21] found that this problem can be avoided if the linear and exponential equations are combined. The result adheres to the high-temperature boundary condition.[20,22] In order to increase the reliability of our assessments, the enthalpies of formation of the all-stoichiometric compounds that appear in the three systems are determined using ab-initio calculations. Thus, the objective target of our work is to critically evaluate all the original experimental data and those derived from ab-initio calculations. Then, we compare the performance of the quadratic (linear + a_0^{Φ}), exponential and combined models to describe the temperature dependence of the interaction energies of the *Ln-Ru* (with $Ln = Nd,\ Gd,$ and Dy) binary systems. Our paper is organized as follows: in Sects. 2, 3 and 4 the bibliographic description details, thermodynamic data used as well as the ab-initio calculations are discussed. In Sects. 5 and 6, the *CALPHAD* assessment of the three systems coupled with the used thermodynamic models are described. Furthermore, in Sect. 7 the modeling results are presented. Finally, the conclusions and most promising prospects are given in Sect. 8.

2 Literature Study on the *Ln-Ru* (with $Ln = Nd,\ Gd,$ and Dy) Phase Diagrams

2.1 *Nd-Ru* Phase Diagram

The phase diagram of the *Nd-Ru* system has been investigated by Palenzona and Canepa[23] using differential thermal analysis, metallographic analysis, x-ray diffraction and electron microscopy in the atomic range of 25-40 at.% *Ru*. The phase diagram has been drawn, mostly taking into account, the thermal effects detected on heating of alloys that were previously prepared, melted in a high-frequency induction furnace, briefly annealed and then subjected to differential thermal analysis. It should be noted that the significant effects of supercooling increase when passing from light to heavier lanthanides. Owing to the high melting temperatures, the *ruthenium*-rich side has not been studied. The invariant peritectic $Liq + (Ru) \leftrightarrow NdRu_2$ temperature is higher than 1873 K, because an alloy of this composition does not melt at this temperature. Moreover, the compounds Nd_3Ru (Fe_3C, type), Nd_5Ru_2, (Mn_5C_2, type), and $NdRu_2$ ($MgCu_2$, type) have been confirmed. Besides, two new intermediate phases have been found: Nd_7Ru_3 which crystallizes with the Sr_7Pt_3 type and $NdRu_x$, at about 38 ± 1 at.% *Ru*, with a structure related to the $Y_{44}Ru_{25}$ type. This compound was reported by Sharifrazi et al.[24] as Nd_5Ru_3 with a complex crystal structure. None of these phases shows appreciable ranges of solid solubility. The addition of *ruthenium* to *neodymium* does not significantly affect the transition temperatures. One can observe that the *La-Ru* and *Nd-Ru* phase diagrams appear to be nearly the same. Moreover, as reported in Fig. 1(a), Okamoto[25] discusses all liquidus boundaries.

2.2 *Dy-Ry* and *Gd-Ru* Phase Diagrams

The *Dy-Ru* and *Gd-Ru* phase diagrams were studied by thermal analysis, metallography, and x-ray phase analysis by Loebich and Raub.[26] Three phases were found to exist in both systems in addition to the already known Laves phase, $LnRu_2$, *i.e.,* Ln_3Ru and Ln_2Ru, which are formed by peritectic reactions, and $Ln_{73}Ru_{27}$, which melts congruently, as $LnRu_2$. The crystal structures of the new phases were not studied in Ref 26. The presence of the $LnRu_2$ Laves phase is also known for other *Ln–Ru* systems (except *Eu-Ru* system) is also known Ref 27, 28. These phases occur in the lighter lanthanides with the cubic $MgCu_2$ type. Those of the heavier lanthanides as well as *Sc* and *Y* crystallize in the hexagonal $MgZn_2$ type. The upper limit of

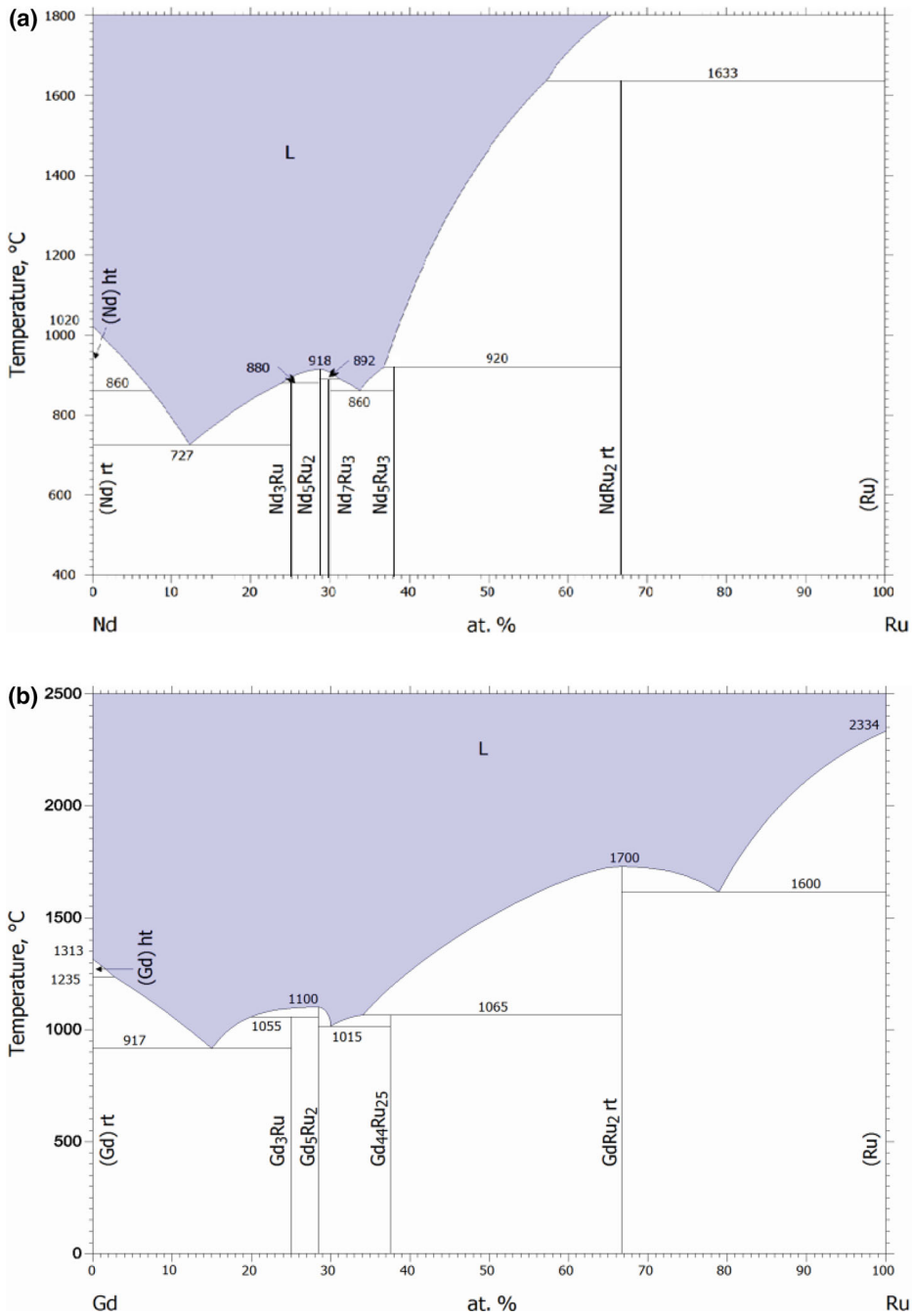
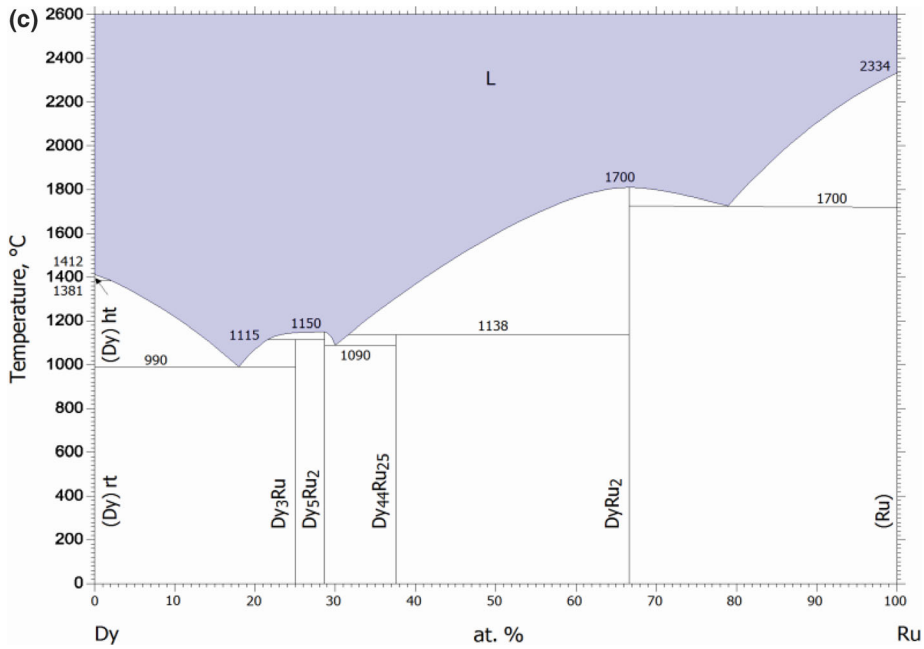


Fig. 1 (a) Experimental *Nd-Ru* phase diagram given by Palenzona and Canepa and revised by Okamoto. Published with permission. Elsevier, Elsevier Science & Technology Journals,[23] The Phase Diagrams of the La-Ru and Nd-Ru Systems, A. Palenzona, F. Canepa, 1990. Springer Nature, Journal of Phase Equalibria,[25] Nd-Ru (Neodymium-Ruthenium), H. Okamoto, 1991. (b) Experimental *Gd-Ru* phase diagram according to Loebich and Raub. Published with permission. Elsevier, Elsevier Science & Technology Journals,[26] Die Legierungen des Rutheniums mit Gadolinium und Dysprosium und ihre magnetischen Eigenschaften, O. Loebich Jr. and E. Raub, 1976. (c) Experimental *Dy-Ru* phase diagram according to Loebich and Raub Published with permission. Elsevier, Elsevier Science & Technology Journals,[26] Die Legierungen des Rutheniums mit Gadolinium und Dysprosium und ihre magnetischen Eigenschaften, O. Loebich Jr. and E. Raub, 1976

temperatures that could be measured with sufficient accuracy of 5 K was 1833 K.[26] It should be emphasized that the melting temperature of congruently melting Laves phases cannot be determined well when the alloys are melted in vacuum since the irreversible loss of lanthanides during evaporation leads to the formation of a *Ru*-rich eutectic melt. The melting temperature of alloys containing 78-80 at.% *Ru* can also be inaccurate. Within the four compounds, only two with the composition $Ln_{73}Ru_{27}$, melt congruently at 1173 K for $Gd_{73}Ru_{27}$ and 1423 K for $Dy_{73}Ru_{27}$. The other two phases are formed peritectically at 1328 K (Gd_3Ru) and 1388 K (Dy_3Ru) and at 1338 K (Gd_2Ru) and 1411 K (Dy_2Ru). The two eutectic points of the *Gd-Ru* system correspond to 1190 K and 1238 K for 70 and 85 at.% *Gd*, and, in the case of the *Dy-Ru* system, they are 1263 K and 1363 K for compositions of 70 and 82 at.% *Dy*. Thus, the temperatures are significantly higher for the *Dy-Ru* alloys. According to the same study, the three phases have certain similarities in both systems. However, no correlation with a known structure could be determined. The mutual solubility of *ruthenium*, *gadolinium* or *dysprosium* has not been established. In Fig. 1(b) and (c), the

Fig. 1 continued



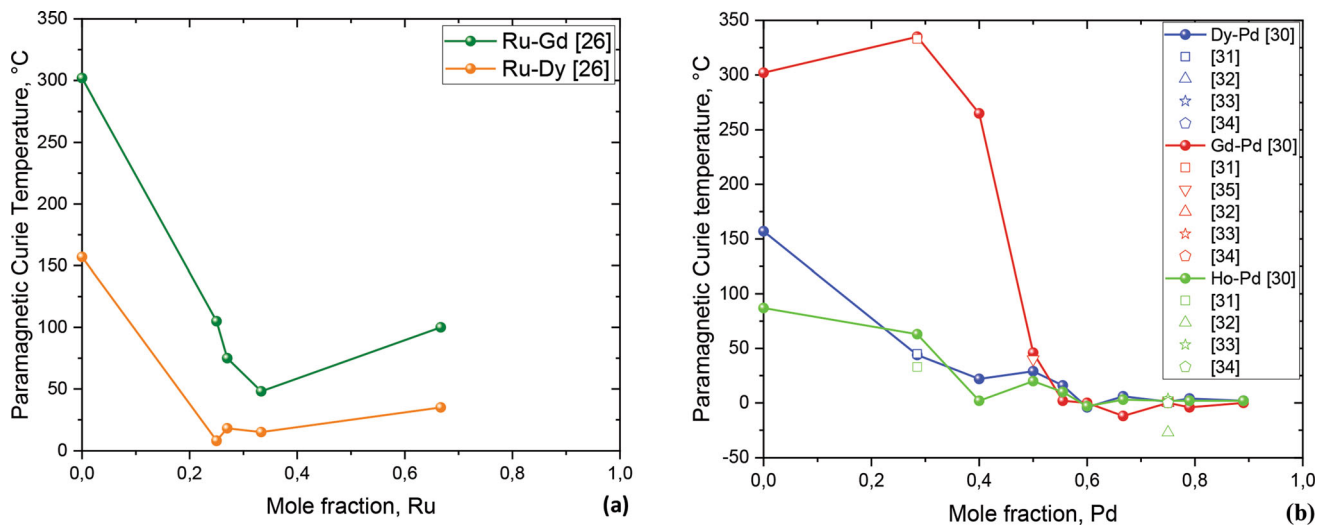


Fig. 2 The Curie temperatures of the *Ln-Ru* (with *Ln* = *Gd* and *Dy*)[26] (a) and *Ln-Pd* (with *Ln* = *Gd*, *Dy*, and *Ho*)[30–35] (b) intermetallic compounds

(Gd,Dy)-Ru phase diagrams are displayed. Also, as shown in Fig. 2(a), the magnetic investigation of the *Gd-Ru* and *Dy-Ru* systems by Ref 26 revealed that the magnetic transition temperatures vary with the composition as in the *Gd-Rh* system.[29] At room temperature, all phases are in the paramagnetic state. The Curie temperature for the $GdRu_2$ which is the Laves phase with a hexagonal structure is 100 K higher than $GdRh_2$ with cubic structure. It is noticeable that in these alloys, the temperature decreases more rapidly with decreasing lanthanide concentration. The same observation was noted for *palladium* alloys,[30–35] as shown in Fig. 2(b). The obtained values for Curie temperature by Ref 30, are negative for $GdPd_2$, $Gd_{0.84}Pd_{3.16}$, Dy_2Pd_3, Ho_2Pd_3, and $HoPd_3$ intermetallic compounds, indicating that the *Pd* moments are aligned antiferromagnetically or ferromagnetically. Crystal structure and lattice parameters data of the stable intermediate binary phases in *Nd-Ru*, *Dy-Ru*, and *Gd-Ru* binary systems are depicted in Table 1.

3 Thermodynamic Data

Due to the experimental difficulties, such as high temperature and oxidation, the thermodynamic data of the *Ln-Ru* (with $Ln = Nd$, Gd, and Dy) binary systems are very limited.

3.1 *Nd-Ru* System

Guo and Kleppa[36] determined the standard enthalpies of formation for Nd_5Ru_2 and $NdRu_2$ intermetallic compounds by direct synthesis calorimetry (*DSC*) at 1477 ± 2 K: ΔH_f (Nd_5Ru_2) = − 17.2 ± 1.9 kJ/mol-at. and ΔH_f $(NdRu_2)$- = − 18.8 ± 1.2 kJ/mol-at. In addition, calorimetric values for $NdRu_2$ were obtained by Shilov et al.[37] from *DTA* data. The data of Shilov et al.[37] (− 19.00 kJ/mol-at.) was obtained at 1030 K. As shown in Table 2, despite the difference between the 298.15 K and 1030 K or 1477 K data, it can be seen that the Shilov value for $NdRu_2$ agrees well with the data obtained by Guo and Kleppa.[36]

3.2 *Gd-Ru* and *Dy-Ru* Systems

Guo and Kleppa[36] also determined the standard enthalpy of formation for $DyRu_2$ at the same temperature (i.e. 1477 ± 2 K): ΔH_f $(DyRu_2)$ = − 27.3 ± 0.9 kJ/mol-at. For the compounds that exist in the *Gd-Ru* system, there is no published measured value of the standard enthalpies of formation. In the three cases, it is necessary to estimate the enthalpies of formation of all compounds that exist in the three systems. For this reason, the Miedema model,[39,40,43] which is a semi-empirical one for predicting thermochemical properties, especially for the enthalpies of formation of compounds and the enthalpies of mixing of solutions, was used. However, the values given by Ref 39, 40, 43 are still significantly more exothermic than the previously measured values, as are our calculated results.

4 Ab-Initio Calculations

In the case of *Nd-Ru* phase diagrams, to supplement the available experimental data, ab-initio calculations based on Density Functional Theory (*DFT*)[49,50] within the plane wave pseudopotentials as implemented in Quantum Espresso code[51] were used to determine the enthalpies of formation of the intermetallic compounds. The exchange correlation interaction was treated within the generalized gradient approximation (*GGA*) given by Perdew, Burke and Ernzerhof (*PBE*).[52] A plane wave cutoff energy of 50 Ry is used for all considered compounds. The ab-initio calculations give the total energy at temperature = 0 K, E^{Φ}. The energy of formation is calculated by the following equation:

$$\Delta_f E = E^{\Phi} - \sum x_i \times E_i^0 \quad \text{(Eq 1)}$$

where E^0 is the total energy of element i in its stable state at 0 K. According to Colinet,[53] the energy of formation corresponds to the enthalpy of formation.

The available experimental, interpolated, ab-initio calculationed data for enthalpies of formation of all solid compounds in the three systems and those presented in the online databases, such as from the Materials project,[41] OQMD,[42] Aflow,[45] and Jarvis,[47] are tabulated in Table 2.

5 Thermodynamic Modeling

The *Ln-Ru* (*Ln* = *Nd, Gd, Dy*) binary systems include the following phases: *Liquid*, (αNd), (βNd), (αDy), (βDy), (αGd), (βGd), and (Ru) solid solution phases, and thirteen stoichiometric compounds:Ln_3Ru, Nd_5Ru_2, Ln_7Ru_3, Nd_5Ru_3, $Ln_{73}Ru_{27}$, Ln_2Ru, and $LnRu_2$.

5.1 Unary Phases

The Gibbs energy function for the element i (i = *Ru*, *Nd*, *Gd*, and *Dy*) is expressed using the following equation:

$$G^{\Phi} - H_i^{SER} = a + b \times T + c \times T \times \ln T + d \times T^2 + e \times T^3 + f \times T^{-1} + g \times T^7 + h \times T^{-9} \quad \text{(Eq 2)}$$

where $H_i^{SER}(298.15K)$ is the molar enthalpy of the element i at 298.15 K and 1.00 bar in its standard element reference (*SER*) state, *hcp_A3* for *Nd, Dy, Gd,* and *Ru*, and T is the absolute temperature; *a, b, c, d, e,* and f are coefficients. The last two terms in (Eq 2) are used for metastable ranges below and above the melting point, g for a liquid below the melting point and h for the solid phases above the melting point. The Gibbs free energy of the element i, in its *SER* state, is denoted by $GHSER_i$ as follows:

$$GHSER_{Ru} = {}^0G_{Ru}^{hcp_A3} = G_{Ru}^{hcp_A3} - H_{Ru}^{hcp_A3}(298.15\text{ K}) \quad \text{(Eq 3)}$$

$$GHSER_{Nd} = {}^0G_{Nd}^{dhcp} = G_{Nd}^{dhcp} - H_{Nd}^{dhcp}(298.15\text{ K}) \quad \text{(Eq 4)}$$

$$GHSER_{Dy} = {}^0G_{Dy}^{hcp_A3} = G_{Dy}^{hcp_A3} - H_{Dy}^{hcp_A3}(298.15\text{ K}) \quad \text{(Eq 5)}$$

Table 1 Crystal structure and lattice parameters data of the stable intermediate binary phases in *Nd-Ru*, *Dy-Ru*, and *Gd-Ru* binary systems

Phase/temperature range (°C)	Composition, at.% *Ru*	Pearson Symbol	Space Group	Strukturbericht designation	Used model	Prototype	Modeling phase	Lattice parameters, nm			References
								a	*b*	*c*	
Nd-Ru binary system											
(βNd) 863–1021	0.00	*cI2*	*P6₃/mmc*	*A2*	*SSM*	*W*	$(Nd,Ru)_1\ (Va)_2$	0.36582	…	…	Ref 63
(αNd) Jusqu'à 863	0.00	*hP4*	*Im $\bar{3}$m*	*A3'*	*SSM*	*αLa*	$(Nd,Ru)_1\ (Va)_3$	0.4130	…	…	
$Nd_3Ru < 880$	25.0	*oP16*	*Pnma*	$D0_{11}$	*SM*	Fe_3C	$(Nd)_3\ (Ru)$	0.7355	0.9675	0.6406	Ref 23, 24
								0.7349	0.9655	0.6399	
								0.7359	0.9650	0.6448	
								0.7353	0.9649	0.6406	
$Nd_5Ru_2 < 915$	28.6	*mC28*	*C2/c*	…	*SM*	…	$(Nd)_5\ (Ru)_2$	1.6295	0.6509@β = 96.67°	0.7342	
								1.6289	0.6505@β = 96.66°	0.7339	
								1.6290	0.6503@β = 96.63°	0.7330	
$Nd_7Ru_3 < 890$	30.0	*oP40*	*Pnma*	…	*SM*	…	$(Nd)_7\ (Ru)_3$	0.7303	2.2830	0.6532	Ref 24
$NdRu_x < 920$	38 ± 1.0	*o***	…	…	*SM*	…	$(Nd)_5\ (Ru)_3$	18.630	1.560	1.560	
$NdRu_2 > 1600$	66.7	*cF24*	*Fd $\bar{3}$m*	*C15*	*SM*	Cu_2Mg	$(Nd)_1(Ru)_2$	0.7614	…	…	Ref 23
								0.7612	…	…	
								0.7637	…	…	
								0.5323[a]		0.9004[a]	
								0.7608	…	…	Ref 24
								0.5293[a]		0.8955[a]	
Dy-Ru binary system											
(βDy) > 1381	0.00	*cI2*	*P6₃/mmc*	*A2*	*SSM*	*W*	$(Nd,Ru)_1\ (Va)_2$	…	…	…	Ref 63
(αDy) 25–1381	0.00	*hP4*	*Im $\bar{3}$m*	*A3'*	*SSM*	*αLa*	$(Nd,Ru)_1\ (Va)_3$	…	…	…	
$Dy_3Ru < 1115$	25.0	…	*Pnma*	…	*SM*	Fe_3C	$(Dy)_3\ (Ru)$	…	…	…	Ref 26
$Dy_{73}Ru_{27} < 1150$	27.0	…	…	…	*SM*	…	$(Dy)_{73}\ (Ru)_{27}$	…	…	…	
$Dy_2Ru < 1138$	33.3	…	*Fd $\bar{3}$m*	…	*SM*	Mg_2Cu	$(Dy)_2(Ru)$	…	…	…	
$DyRu_2 > 1700$	66.7	…	*Fd $\bar{3}$m*	…	*SM*		$(Dy)(Ru)_2$	…	…	…	
Gd-Ru binary system											
(βGd) > 1381	0.00	*cI2*	*P6₃/mmc*	*A2*	*SSM*	*W*	$(Gd,Ru)_1\ (Va)_2$	…	…	…	Ref 63
(αGd) 25–1381	0.00	*hP4*	*Im $\bar{3}$m*	*A3'*	*SSM*	*αLa*	$(Gd,Ru)_1\ (Va)_3$	…	…	…	
$Gd_3Ru < 1055$	25.0	…	*Pnma*		*SM*	Ni_3Sn	$(Gd)_3\ (Ru)$	…	…	…	Ref 26
$Gd_{73}Ru_{27} < 1100$	27.0	…	…	…	*SM*	…	$(Gd)_{73}\ (Ru)_{27}$	…	…	…	
$Gd_2Ru < 1065$	33.3	…	*Cmcm*	…	*SM*		$(Gd)_2(Ru)$	…	…	…	
$GdRu_2 > 1700$	0.667	…	*P6₃/mmc*	…	*SM*	$MgZn_2$	$(Gd)(Ru)_2$	…	…	…	
(Ru) < 2334	100	*hP2*	*P6₃/mmc*	*A3*	*SSM*	*Mg*	$(Ln,Ru)_1\ (Va)_3$	0.27058	…	0.42816	Ref 63
		cF4	*Fm $\bar{3}$m*	*A1*		*Cu*					

SSM Subregular solution model, *SM* Sublattice model.

[a]High pressure phase.

Table 2 The comparison between the optimized, calculated, predicted, interpolated, and measured enthalpies, entropies, and Gibbs energies of formation of the thirteen solid compounds that exist in the *Nd-Ru*, *Dy-Ru*, and *Gd-Ru* systems

Compounds	*Ln*	ΔH_f (kJ/mol-at.)	ΔS_f (J/mol-at K)	ΔG_f at 1000 K (kJ/mol-at)	Method used	References
Ln_3Ru	*La*	− 15.002	− 8.260	− 6.742	Optimization	Ref 38
	Nd	− 15.080	− 1.222	− 13.858	Optimization^{QM}	Current work
		− 15.183	− 0.002	− 15.181	Optimization^{EM}	
		− 16.974	− 2.023	− 14.950	Optimization^{CM}	
		− 25.095	…	…	Miedema's model	Ref 39, 40
		− 21.451	…	…	Ab-initio	Current work
		− 18.790	…	…	Ab-initio	Ref 41
		− 18.117	…	…	Ab-initio	Ref 42
		− 21.846	…	…	Modified Miedema's model	Ref 43
	Gd	− 21.447	− 0.139	− 21.307	Optimization^{QM}	Current work
		− 15.612	− 0.056	− 15.555	Optimization^{EM}	
		− 15.131	− 0.329	− 14.802	Optimization^{CM}	
		− 26.599	…	…	Miedema's model	Ref 39, 40
		− 23.258	…	…	Modified Miedema's model	Ref 43
		− 27.175	…	…	Ab-initio	Ref 41
	Dy	− 16.545	− 0.138	− 16.407	Optimization^{QM}	Current work
		− 15.693	− 0.055	− 15.637	Optimization^{EM}	
		− 15.978	− 0.166	− 15.811	Optimization^{CM}	
		− 28.644	…	…	Miedema's model	Ref 39, 40
		− 25.204	…	…	Modified Miedema's model	Ref 43
		− 31.223	…	…	Ab-initio	Ref 41
		− 30.934	…	…	Ab-initio	Ref 42
Ln_7Ru_3	*La*	− 16.519	− 8.620	− 07.899	Optimization	Ref 44
	Nd	− 15.118	− 0.124	− 14.994	Optimization^{QM}	Current work
		− 16.331	− 0.120	− 16.211	Optimization^{EM}	
		− 17.529	− 1.372	− 16.157	Optimization^{CM}	
		− 30.181	…	…	Miedema's model	Ref 39, 40
		− 25.145	…	…	Ab-initio (Magnetic)	Current work
		− 25.367	…	…	Modified Miedema's model	Ref 43
$LnRu_2$	*La*	− 13.801	− 5.880	− 07.921	Optimization	Ref 38
		− 10.3 ± 2.8	…	…	DSC	Ref 44
		− 11	…	…		Ref 37
	Nd	− 16.185	− 0.024	− 16.161	Optimization^{QM}	Current work
		− 16.650	− 2.300	− 14.349	Optimization^{EM}	
		− 21.949	− 5.001	− 16.947	Optimization^{CM}	

Table 2 continued

Compounds	*Ln*	ΔH_f (kJ/mol-at.)	ΔS_f (J/mol-at K)	ΔG_f at 1000 K (kJ/mol-at)	Method used	References
		− 18.8 ± 1.2	…	…	DSC at 1477 ± 2 K	Ref 36
		− 18.6 ± 1.0	…	…		
		− 19.000	…	…	DSC	Ref 37
		− 43.763	…	…	Miedema's model	Ref 39, 40
		− 33.894	…	…	Ab-initio	Current work
		− 15.032	…	…	Ab-initio	Ref 41
		− 13.104	…	…		
		− 15.723	…	…	Ab-initio	Ref 45
		− 34.129	…	…	Modified Miedema's model	Ref 43
	Gd	− 40.312	− 0.095	− 40.217	Optimization^{QM}	Current work
		− 26.827	− 0.869	− 25.958	Optimization^{EM}	
		− 26.110	− 0.414	− 25.696	Optimization^{CM}	
		− 45.670	…	…	Miedema's model	Ref 39, 40
		− 35.999	…	…	Modified Miedema's model	Ref 43
		− 24.681	…	…	Ab-initio	Ref 41
		− 26.308	…	…	Ab-initio	Ref 45
		− 23.800	…	…		
	Dy	− 30.387	− 0.095	− 30.292	Optimization^{QM}	Current work
		− 27.060	− 0.869	− 26.191	Optimization^{EM}	
		− 29.491	− 0.372	− 29.118	Optimization^{CM}	
		− 48.108	…	…	Miedema's model	Ref 39, 40
		− 27.300	…	…	DSC at 1477 ± 2 K	Ref 46
		− 38.476	…	…	Modified Miedema's model	Ref 43
		− 27.753	…	…	Ab-initio	Ref 41
Ln_5Ru_2	*La*	− 16.121	− 8.620	− 07.501	Optimization	Ref 38
	Nd	− 15.088	− 0.238	− 14.849	Optimization^{QM}	Current work
		− 16.261	− 0.128	− 16.132	Optimization^{EM}	
		− 17.376	− 1.386	− 15.989	Optimization^{CM}	
		− 17.2 ± 1.9	…	…	DSC at 1477 ± 2 K	Ref 36
		− 28.735	…	…	Miedema's model	Ref 39, 40
		− 24.125	…	…	Ab-initio	Current work
		− 21.102	…	…	Ab-initio	Ref 41
		− 20.524	…	…	Ab-initio	Ref 47
		− 24.403			Modified Miedema's model	Ref 43

Table 2 continued

Compounds	*Ln*	ΔH_f (kJ/mol-at.)	ΔS_f (J/mol-at K)	ΔG_f at 1000 K (kJ/mol-at)	Method used	References
Ln_5Ru_3	*La*	− 17.158	− 9.320	− 07.838	Optimization	Ref 48
	Nd	− 15.289	0.363	− 15.653	Optimization^{QM}	Current work
		− 16.342	− 0.018	− 16.323	Optimization^{EM}	
		− 18.370	− 1.487	− 16.882	Optimization^{CM}	
		− 37.442	…	…	Miedema's model	Ref 39, 40
		− 33.870	…	…	Ab-initio	Current work
		− 13.779	…	…	Ab-initio	Ref 42
		− 11.755	…	…		
		− 29.667			Modified Miedema's model	Ref 43
	Gd	− 22.787	− 0.064	− 22.723	Optimization^{QM}	Current work
		− 16.380	− 0.007	− 16.373	Optimization^{EM}	
		− 15.626	− 0.017	− 15.609	Optimization^{CM}	
		− 28.761	…		Miedema's model	Ref 39, 40
		− 24.820	…		Modified Miedema's model	Ref 43
	Dy	− 17.477	− 0.064	− 17.413	Optimization^{QM}	Current work
		− 16.483	− 0.007	− 16.476	Optimization^{EM}	
		− 16.740	− 0.013	− 16.726	Optimization^{CM}	
		− 30.969	…		Miedema's model	Ref 39, 40
		− 29.913	…		Modified Miedema's model	Ref 43
		− 34.403	…	…	Ab-initio	Ref 41
		− 34.114	…	…	Ab-initio	Ref 42
Ln_2Ru	*Gd*	− 25.661	− 0.071	− 25.590	Optimization^{QM}	Current work
		− 18.032	− 0.082	− 17.949	Optimization^{EM}	
		− 17.284	− 0.014	− 17.269	Optimization^{CM}	
		− 35.480	…	…	Miedema's model	Ref 39, 40
		− 29.280	…	…	Modified Miedema's model	Ref 43
		− 20.622	…	…	Ab-initio	Ref 41
	Dy	− 19.547	− 0.071	− 19.476	Optimization^{QM}	Current work
		− 18.155	− 0.078	− 18.077	Optimization^{EM}	
		− 18.759	− 0.023	− 18.735	Optimization^{CM}	
		− 38.167	…	…	Miedema's model	Ref 39, 40
		− 31.800	…	…	Modified Miedema's model	Ref 43

^{QM}: Quadratic model (= Linear model + $d*T^2$).

^{EM}: Exponential mode.

^{CM}: Combined model.

$$GHSER_{Gd} = {}^{0}G_{Gd}^{hcp_A3} = G_{Gd}^{hcp_A3} - H_{Gd}^{hcp_A3}(298.15\ \mathrm{K}) \quad \text{(Eq 6)}$$

In the present work, the Gibbs energies functions are taken from the *SGTE* (Scientific Group Thermodata Europe) database *PURE* (1, 3, and 4) from T = 298.15 to 10,000 K compiled by Dinsdale.[54]

5.2 Solution Phases

The molar Gibbs energy of the liquid and solid solution phases, $(\alpha Nd), (\beta Nd), (\alpha Dy), (\beta Dy)$, $(\alpha Gd), (\beta Gd)$, and (Ru) can be expressed as the sum of the reference part, ideal part, excess part, and magnetic part:

$$G_m^{\Phi} = {}^{ref}G^{\Phi} + {}^{id}G^{\Phi} + {}^{ex}G^{\Phi} + {}^{mag}G^{\Phi} \quad \text{(Eq 7)}$$

$$G_m^{\Phi} = x_{Ln}^{\Phi} \times {}^{0}G_{Ln}^{\Phi} + x_{Ru}^{\Phi} \times {}^{0}G_{Ru}^{\Phi} + R \times T \times \left(x_{Ln}^{\Phi} \times \ln x_{Ln}^{\Phi} + x_{Ru}^{\Phi} \times \ln x_{Ru}^{\Phi}\right) + {}^{ex}G^{\Phi} + {}^{mag}G^{\Phi} \quad \text{(Eq 8)}$$

where R = 8.31447 J/mol K is the universal gas constant, TK is the absolute temperature, x_{Ln} and x_{Ru} are the mole fraction of the elements $Ln = Nd$, Gd, and Dy and Ru, respectively. The unit "mol" refer to one mole of atoms.

The symbol ${}^{mag}G^{\Phi}$ describes the magnetic contribution to the molar Gibbs energy.[55] For all systems, during the optimization procedure, we did not take into account this contribution to the Gibbs energy of the intermetallic compounds and solution phases for the following reasons: the Curie temperature of all the compounds in the *Ln-Ru* binary systems is far below 298.15 K, the compounds are in a paramagnetic state above this temperature and the thermodynamic functions of the *Ln* and *Ru* elements are given in a temperature range above 298.15 K[54] (see Fig. 2(a) for *Gd-Ru* and *Dy-Ru* systems).

The excess Gibbs energy, ${}^{ex}G_m^{\alpha}$ is described by using a Redlich–Kister type polynomial representation,[10,11,54,56] i.e.

$${}^{ex}G^{\Phi} = x_{Ru}^{\Phi} \times \left(1 - x_{Ru}^{\Phi}\right) \times \sum_{j=0} L_j^{\Phi} \times \left(1 - 2 \times x_{Ru}^{\Phi}\right)^j \quad \text{(Eq 9)}$$

where the interaction energies of j-th order (usually j = 0,1,2) for the binary *Ln-Ru* solutions in phase Φ, L_j^{Φ} (J/mol-at.) are taken to vary linearly with temperature:

$$L_j^{\Phi} = a_j^{\Phi} + b_j^{\Phi} \times T \ (j = 0, 1, 2, \ldots) \quad \text{(Eq 10)}$$

with the excess entropy $s = -b_j^{\Phi}$ (J/mol.K) and the enthalpy $h = a_j^{\Phi}$ (J/mol) for the *j-th* order interaction energy of the *Ln-Ru* solutions in phase Φ.

Then, Kaptay[18,57] proposed using function to describe the excess Gibbs free energy of the solution and liquid phases. The binary interaction parameter L_j^{Φ}, assessed in the present work, is temperature dependent, as follows:

$$L_j^{\Phi} = h_j^{\Phi} \times \exp\left(-\frac{T}{\tau_j^{\Phi}}\right) \quad \text{(Eq 11)}$$

where h_j^{Φ} (J/mol-at.): The enthalpy part of the interaction energy is extrapolated to T = 0 K. τ_j^{Φ}: The ratio of the heat of the mixing parameter and the excess entropy parameter; thus, it is a special temperature, at which the L_0 parameter crosses zero and changes its sign if it were described by the quadratic model with $d_0^{\Phi} = 0$ (thus τ_j^{Φ} must be always positive).

By identifying h_j^{Φ} with a_j^{Φ} and τ_j^{Φ} with a_j^{Φ}/b_j^{Φ}, Eq 11 becomes Eq 12 with the excess entropy $b_j^{\Phi} = -s$:

$$L_j^{\Phi} = a_j^{\Phi} \times exp\left(\left(-\frac{b_j^{\Phi}}{a_j^{\Phi}}\right) \times T\right) \quad \text{(Eq 12)}$$

According to Ref 18, 21, the linear (quadratic with $d_0^{\Phi} = 0$) and exponential equations are combined as:

$$L_j^{\Phi} = \left(h_j^{\Phi} - T \times s_j^{\Phi}\right) \times exp\left(-\frac{T}{\tau_j^{\Phi}}\right) \quad \text{(Eq 13)}$$

where s_j^{Φ} (J/mol K) is the entropy part of the interaction energy extrapolated to T = 0 K.

In the current work, we are comparing the performances of the quadratic, exponential and combined models to describe the temperature dependence of the interaction energies. We have limited the leading term of the interaction energy to j = 0. Thus, the excess Gibbs energy of the same solution Φ will be compared in three forms:

$$\text{The quadratic model}: {}^{ex}G^{\Phi} = x \times (1 - x) \times \left(h_0^{\Phi} - T \times s_0^{\Phi} + d_0^{\Phi} \times T^2\right) \quad \text{(Eq 14)}$$

$$\text{The exponential model}: {}^{ex}G^{\Phi} = x \times (1 - x) \times h_{e,0}^{\Phi} \times exp\left(-\frac{T}{\tau_{e,0}^{\Phi}}\right) \quad \text{(Eq 15)}$$

$$\text{The combined model}: {}^{ex}G^{\Phi} = x \times (1 - x) \times \left(h_{c,0}^{\Phi} - T \times s_{0,c}^{\Phi}\right) \times exp\left(-\frac{T}{\tau_{0,c}^{\Phi}}\right) \quad \text{(Eq 16)}$$

where h_0^{Φ} (J/mol) and s_0^{Φ} (J/mol K) are the temperature independent enthalpy and entropy terms of the quadratic

model, $h^{\Phi}_{e,0}$ (J/mol) and $\tau^{\Phi}_{e,0}$ (K) are the zero-Kelvin enthalpy term and the special temperature of the exponential model, while $h^{\Phi}_{c,0}$ (J/mol), $s^{\Phi}_{e,0}$ (J/mol K), and $\tau^{\Phi}_{c,0}$ (K) are the zero-Kelvin enthalpy terms, the zero-Kelvin entropy term and the special temperature of the combined model, respectively.

The artifact liquid solution re-stabilization at high temperatures takes place if $d^2\Delta G/dx^2 < 0$ thermodynamic condition is satisfied at any composition. Equations for the heat of mixing, excess entropy of mixing and excess heat capacity of mixing will be derived by the following valid equations:

$$S^E_{\Phi} = -\frac{dG^E_{\Phi}}{dT} \quad \text{(Eq 17)}$$

$$H_{\Phi} = G^E_{\Phi} - T \times \frac{dG^E_{\Phi}}{dT} \quad \text{(Eq 18)}$$

$$\Delta C^E_{p,\Phi} = -T \times \frac{d^2G^E_{\Phi}}{dT^2} \quad \text{(Eq 19)}$$

The excess term of the Gibbs energy was optimized for terminal solid solutions to describe the solubility of *Ln* in *Ru*. To correctly describe the homogeneity range of *hcp_A3 (Ru)* phase, ${}^0L$ was used. In the case of the quadratic model, for the liquid phase the ${}^2L$ term was necessary to reproduce the low-melting eutectic (at temperature of 1093 K) in the *Ln*-rich part of the *Ln-Ru* phase diagram.

5.3 Stoichiometric Compounds

Due to their negligible homogeneity ranges, all intermediate compounds Ln_3Ru, Nd_5Ru_2, Ln_7Ru_3, Nd_5Ru_3, $Ln_{73}Ru_{27}$, Ln_2Ru, and $LnRu_2$ that exist in *Ln-Ru* binary systems and are treated in our optimization as stoichiometric phases and described as A_pB_q. The Gibbs energy of each phase is denoted as ${}^0G_{Ln_pRu_q}$ and described using the following formula:

$${}^0G_{Ln_pRu_q} = \frac{p}{p+q} \times {}^0G_{Ln} + \frac{q}{p+q} \times {}^0G_{Ru} + \Delta G_f^{Ln_pRu_q} \quad \text{(Eq 20)}$$

where ${}^0G_{Ln}$ and ${}^0G_{Ru}$ is the Gibbs energy of the pure elements *cerium*, *neodymium*, and *ruthenium*, respectively; $\Delta G_f^{Ln_pRu_q}$ is the Gibbs energy of formation per mole of formula units of Ln_pRu_q. Because of the lack of experimental measurements, it is assumed that the Neumann–Kopp rule applies to the heat capacity, i.e. $\Delta C_P = 0$. Thus $\Delta G_f^{Ln_pRu_q}$ can be given by the following expression:

$$\Delta G_f^{Ln_pRu_q} = a + b \times T \quad \text{(Eq 21)}$$

where the parameters *a* and *b* were to be optimized for each compound in the current work. The coefficient *a* can be estimated according to the corresponding experimental enthalpies of formation, and then evaluated by taking into account both phase equilibria and thermodynamic data.

6 Optimization Procedure

In this section, we describe the approach to the thermodynamic modelling of the studied systems. As the first step, we started with the liquid phase, using the zero order interaction parameters. Subsequently, intermetallic compounds were introduced into the modeling one by one. We first introduced the thermodynamic parameters of the $LnRu_2$ and Nd_5Ru_2 optimized compounds based on the enthalpy of formation data measured by Selhaoui and Kleppa[58] and Guo and Kleppa[36,46] using direct synthetic calorimetry at high temperature, respectively. During the course of the optimization, the first and second order interaction parameters of the liquid phase were introduced to describe the shape of liquidus in the phase diagrams. The calculated invariant temperatures were adjusted to minimize the square sum of the errors between the experimental data and the computed values until most of the selected experimental information was reproduced within the expected uncertainty limits. The weights were adjusted systematically to improve the overall consistency of the data. The final thermodynamic parameters obtained are listed in Table 3.

7 Results and Discussion

The thirteen intermediate phases Ln_3Ru, Nd_5Ru_2, Ln_7Ru_3, Nd_5Ru_3, $Ln_{73}Ru_{27}$, Ln_2Ru, and $LnRu_2$ in the *Ln-Ru* binary systems have been treated as stoichiometric compounds while a solution model has been used for the description of the liquid phase and (αNd), (βNd), (αDy), (βDy), (αGd), (βGd), and (Ru) for solid solutions. The excess term of the Gibbs energy of the liquid and solution phases was assessed with three models: (a) quadratic model,[9–14,56] (b) exponential model,[18,57] and (c) combined model.[21] A detailed analysis is given below.

7.1 *Nd-Ru* Binary System

The superimposed calculated *Nd-Ru* phase diagram with the optimized thermodynamic parameters obtained using the three models is depicted in Fig. 3(a). This diagram is very similar to the experimental one established by

Table 3 Thermodynamic parameters in the *Nd-Ru*, *Dy-Ru*, and *Gd-Ru* binary systems

Phase/temperature range (°C)	Thermodynamic model	Used model**	Thermodynamic parameters[a]
Nd-Ru binary system			
Liquid	$(Nd,Ru)_1$	QM	${}^0L^{Liq}_{Nd,Ru} = -117571.854 + 68.126136 \times T - 0.0189 \times T^2$
			${}^1L^{Liq}_{Nd,Ru} = -41946.682 + 13.0198 \times T$
			${}^2L^{Liq}_{Nd,Ru} = 57601.963 - 24.26049 \times T$
		EM	${}^0L^{Liq}_{Nd,Ru} = -131438.590 \times exp(-0.000653 \times T)$
			${}^1L^{Liq}_{Nd,Ru} = -147721.876 \times exp(-0.0009757 \times T)$
			${}^2L^{Liq}_{Nd,Ru} = +5971587 \times exp(-0.00460849 \times T)$
		CM	${}^0L^{Liq}_{Nd,Ru} = (-141170.126 - 16.693 \times T) \times exp(-0.000766 \times T)$
			${}^1L^{Liq}_{Nd,Ru} = (-67436.733 + 3.3669 \times T) \times exp(-0.0005139 \times T)$
			${}^2L^{Liq}_{Nd,Ru} = (69389.06 - 35.277 \times T) \times exp(-0.00260607 \times T)$
(βNd) 863–1021	$(Nd,Ru)_1(Va)_3$*	QM	No excess term optimized
		EM	
		CM	
(αNd) < 863	$(Nd,Ru)_1(Va)_{0.5}$*	QM	
		EM	
		CM	
(Ru) < 2334	$(Nd,Ru)_1(Va)_{0.5}$*	QM	${}^0L^{HCP_A3}_{Nd,Ru} = +25000$
		EM	${}^0L^{HCP_A3}_{Nd,Ru} = 229908.01 \times exp(-0.000844 \times T)$
		CM	${}^0L^{HCP_A3}_{Nd,Ru} = 53128.175 \times exp(-0.000103 \times T)$
Nd_3Ru < 880	$Nd_{0.750}$:$Ru_{0.250}$	QM	${}^0G^{Nd_3Ru}_{Nd:Ru} - 0.750 \times {}^{298.15}H^{dhcp}_{Nd} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.750 \times {}^0G^{dhcp}_{Nd} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -15080.37 + 1.221 \times T$
		EM	${}^0G^{Nd_3Ru}_{Nd:Ru} - 0.750 \times {}^{298.15}H^{dhcp}_{Nd} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.750 \times {}^0G^{dhcp}_{Nd} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -15183.493 + 0.00162 \times T$
		CM	${}^0G^{Nd_3Ru}_{Nd:Ru} - 0.750 \times {}^{298.15}H^{dhcp}_{Nd} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.750 \times {}^0G^{dhcp}_{Nd} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -16974.075 + 2.023 \times T$
Nd_5Ru_2 < 915	$Nd_{0.714}$:$Ru_{0.286}$	QM	${}^0G^{Nd_5Ru_2}_{Nd:Ru} - 0.714 \times {}^{298.15}H^{dhcp}_{Nd} - 0.286 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.714 \times {}^0G^{dhcp}_{Nd} + 0.286 \times {}^0G^{hcp_A3}_{Ru} = -15088.571 + 0.238 \times T$
		EM	${}^0G^{Nd_5Ru_2}_{Nd:Ru} - 0.714 \times {}^{298.15}H^{dhcp}_{Nd} - 0.286 \times {}^{298.15}H^{hcp_A3}_{Ru} =$
			$+0.714 \times {}^0G^{dhcp}_{Nd} + 0.286 \times {}^0G^{hcp_A3}_{Ru} = -16261.386 + 0.128 \times T$
		CM	${}^0G^{Nd_5Ru_2}_{Nd:Ru} - 0.714 \times {}^{298.15}H^{dhcp}_{Nd} - 0.286 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.714 \times {}^0G^{dhcp}_{Nd} + 0.286 \times {}^0G^{hcp_A3}_{Ru} = -17376.185 + 1.386 \times T$

Table 3 continued

Phase/temperature range (°C)	Thermodynamic model	Used model**	Thermodynamic parameters[a]
Nd_7Ru_3 < 890	*$Nd_{0.700}$:$Ru_{0.300}$*	QM	${}^0G^{Nd_7Ru_3}_{Nd:Ru} - 0.700 \times {}^{298.15}H^{dhcp}_{Nd} - 0.300 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.700 \times {}^0G^{dhcp}_{Nd} + 0.300 \times {}^0G^{hcp_A3}_{Ru} = -15118.218 + 0.124 \times T$
		EM	${}^0G^{Nd_7Ru_3}_{Nd:Ru} - 0.700 \times {}^{298.15}H^{dhcp}_{Nd} - 0.300 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.700 \times {}^0G^{dhcp}_{Nd} + 0.300 \times {}^0G^{hcp_A3}_{Ru} = -16331.8 + 0.120 \times T$
		CM	${}^0G^{Nd_7Ru_3}_{Nd:Ru} - 0.700 \times {}^{298.15}H^{dhcp}_{Nd} - 0.300 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.700 \times {}^0G^{dhcp}_{Nd} + 0.300 \times {}^0G^{hcp_A3}_{Ru} = -17529.499 + 1.372 \times T$
Nd_5Ru_3 < 920	*$Nd_{0.625}$:$Ru_{0.375}$*	QM	${}^0G^{Nd_5Ru_3}_{Nd:Ru} - 0.625 \times {}^{298.15}H^{dhcp}_{Nd} - 0.375 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.625 \times {}^0G^{dhcp}_{Nd} + 0.375 \times {}^0G^{hcp_A3}_{Ru} = -15289.66 - 0.363 \times T$
		EM	${}^0G^{Nd_5Ru_3}_{Nd:Ru} - 0.625 \times {}^{298.15}H^{dhcp}_{Nd} - 0.375 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.625 \times {}^0G^{dhcp}_{Nd} + 0.375 \times {}^0G^{hcp_A3}_{Ru} = -16342.206 - 0.0189 \times T$
		CM	${}^0G^{Nd_5Ru_3}_{Nd:Ru} - 0.625 \times {}^{298.15}H^{dhcp}_{Nd} - 0.375 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.625 \times {}^0G^{dhcp}_{Nd} + 0.375 \times {}^0G^{hcp_A3}_{Ru} = -18370.163 + 1.487 \times T$
$NdRu_2$ > 1600	*$Nd_{0.333}$:$Ru_{0.667}$*	QM	${}^0G^{NdRu_2}_{Nd:Ru} - 0.333 \times {}^{298.15}H^{dhcp}_{Nd} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.333 \times {}^0G^{dhcp}_{Nd} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -16185.701 + 0.0242 \times T$
		EM	${}^0G^{NdRu_2}_{Nd:Ru} - 0.333 \times {}^{298.15}H^{dhcp}_{Nd} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.333 \times {}^0G^{dhcp}_{Nd} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -16650.24 + 2.3 \times T$
		CM	${}^0G^{NdRu_2}_{Nd:Ru} - 0.333 \times {}^{298.15}H^{dhcp}_{Nd} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.333 \times {}^0G^{dhcp}_{Nd} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -21949.579 + 5.001 \times T$
Dy-Ru binary system			
Liquid	*(Dy,Ru)$_1$*	QM	${}^0L^{Liq}_{Dy,Ru} = -114253.113 + 46.898 \times T - 0.0178 \times T^2$
			${}^1L^{Liq}_{Dy,Ru} = -88812.948 + 59.41 \times T$
			${}^2L^{Liq}_{Dy,Ru} = 14383.368 - 0.03809 \times T$
		EM	${}^0L^{Liq}_{Dy,Ru} = -77438.859 \times exp(-0.00000273 \times T)$
			${}^1L^{Liq}_{Dy,Ru} = +3175.91 \times exp(-0.000407 \times T)$
			${}^2L^{Liq}_{Dy,Ru} = -33065.42 \times exp(-0.00045 \times T)$
		CM	${}^0L^{Liq}_{Dy,Ru} = (-66429.93 - 9.881 \times T) \times exp(-0.00000112 \times T)$
			${}^1L^{Liq}_{Dy,Ru} = (-147081.198 + 100 \times T) \times exp(-0.000254 \times T)$
			${}^2L^{Liq}_{Dy,Ru} = (7342.412 + 5.687 \times T) \times exp(-0.0000564 \times T)$
(βDy) > 1381	*(Dy,Ru)$_1$(Va)*$_3$*	QM	No excess term optimized
		EM	
		CM	

Table 3 continued

Phase/temperature range (°C)	Thermodynamic model	Used model**	Thermodynamic parameters[a]
(αDy) 25–1381	$(Dy,Ru)_1(Va)_{0.5}$*	QM	${}^0L^{HCP_A3}_{Dy,Ru} = 24132.3 - 0.000303 \times T$
		EM	${}^0L^{HCP_A3}_{Dy,Ru} = 25000 \times exp(-0.0001308 \times T)$
		CM	${}^0L^{HCP_A3}_{Dy,Ru} = 25000 \times exp(-0.00014 \times T)$
(Ru) < 2334	$(Dy,Ru)_1(Va)_{0.5}$*	QM	${}^0L^{HCP_A3}_{Dy,Ru} = 24132.3 - 0.000303 \times T$
		EM	${}^0L^{HCP_A3}_{Dy,Ru} = 25000 \times exp(-0.0001308 \times T)$
		CM	${}^0L^{HCP_A3}_{Dy,Ru} = 25000 \times exp(-0.00014 \times T)$
Dy_3Ru < 1115	$Dy_{0.750}$:$Ru_{0.250}$	QM	${}^0G^{Dy_3Ru}_{Dy:Ru} - 0.750 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.750 \times {}^0G^{hcp_A3}_{Dy} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -16545.69 + 0.138 \times T$
		EM	${}^0G^{Dy_3Ru}_{Dy:Ru} - 0.750 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.750 \times {}^0G^{hcp_A3}_{Dy} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -15693.75 + 0.056 \times T$
		CM	${}^0G^{Dy_3Ru}_{Dy:Ru} - 0.750 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.750 \times {}^0G^{hcp_A3}_{Dy} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -15978.22 + 0.166 \times T$
$Dy_{73}Ru_{27}$ < 1150	$Dy_{0.730}$:$Ru_{0.270}$	QM	${}^0G^{Dy_{73}Ru_{27}}_{Dy:Ru} - 0.730 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.270 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.730 \times {}^0G^{hcp_A3}_{Dy} + 0.270 \times {}^0G^{hcp_A3}_{Ru} = -17477.57 + 0.064 \times T$
		EM	${}^0G^{Dy_{73}Ru_{27}}_{Dy:Ru} - 0.730 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.270 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.730 \times {}^0G^{hcp_A3}_{Dy} + 0.270 \times {}^0G^{hcp_A3}_{Ru} = -16483.566 + 0.00701 \times T$
		CM	${}^0G^{Dy_{73}Ru_{27}}_{Dy:Ru} - 0.730 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.270 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.730 \times {}^0G^{hcp_A3}_{Dy} + 0.270 \times {}^0G^{hcp_A3}_{Ru} = -15978.22 + 0.0139 \times T$
Dy_2Ru < 1138	$Dy_{0.667}$:$Ru_{0.333}$	QM	${}^0G^{Dy_2Ru}_{Dy:Ru} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.667 \times {}^0G^{hcp_A3}_{Dy} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -19547.97 + 0.0715 \times T$
		EM	${}^0G^{Dy_2Ru}_{Dy:Ru} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.667 \times {}^0G^{hcp_A3}_{Dy} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -18155.539 + 0.0784 \times T$
		CM	${}^0G^{Dy_2Ru}_{Dy:Ru} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.667 \times {}^0G^{hcp_A3}_{Dy} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -18759.1552 + 0.0239 \times T$
$DyRu_2$ > 1700	$Dy_{0.333}$:$Ru_{0.667}$	QM	${}^0G^{DyRu_2}_{Dy:Ru} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.333 \times {}^0G^{hcp_A3}_{Dy} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -30387.76 + 0.0953 \times T$
		EM	${}^0G^{DyRu_2}_{Dy:Ru} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.333 \times {}^0G^{hcp_A3}_{Dy} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -27060.854 + 0.869 \times T$
		CM	${}^0G^{DyRu_2}_{Dy:Ru} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Dy} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.333 \times {}^0G^{hcp_A3}_{Dy} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -29491.704 + 0.372 \times T$

Table 3 continued

Phase/temperature range (°C)	Thermodynamic model	Used model**	Thermodynamic parameters[a]
Gd-Ru binary system			
Liquid	*(Gd,Ru)$_1$*	QM	${}^0L^{Liq}_{Gd,Ru} = -164983.881 + 74.404 \times T - 0.0288 \times T^2$
			${}^1L^{Liq}_{Gd,Ru} = -82448.321 + 64.61 \times T$
			${}^2L^{Liq}_{Gd,Ru} = 25316.165 - 0.0347 \times T$
		EM	${}^0L^{Liq}_{Gd,Ru} = -101278.46 \times exp(-0.000154 \times T)$
			${}^1L^{Liq}_{Gd,Ru} = +9683.226 \times exp(-0.0002781 \times T)$
			${}^2L^{Liq}_{Gd,Ru} = -35793.821 \times exp(-0.00073 \times T)$
		CM	${}^0L^{Liq}_{Gd,Ru} = (-103857.405 + 0.779 \times T) \times exp(-0.000203 \times T)$
			${}^1L^{Liq}_{Gd,Ru} = (-139775.25 + 106.93 \times T) \times exp(-0.000313 \times T)$
			${}^2L^{Liq}_{Gd,Ru} = (178337.031 - 13.809 \times T) \times exp(-0.00186 \times T)$
(βGd) > 1381	*(Gd,Ru)$_1$(Va)$_3$**	QM	No excess term optimized
		EM	
		CM	
(αGd) 25–1381	*(Gd,Ru)$_1$(Va)$_{0.5}$**	QM	${}^0L^{hcp_A3}_{Gd,Ru} = 24245.92$
		EM	${}^0L^{hcp_A3}_{Gd,Ru} = +25000 \times exp(-0.00013 \times T)$
		CM	${}^0L^{hcp_A3}_{Gd,Ru} = +25002.5 \times exp(-0.000323 \times T)$
(Ru) < 2334	*(Gd,Ru)$_1$(Va)$_{0.5}$**	QM	${}^0L^{hcp_A3}_{Gd,Ru} = 24245.92$
		EM	${}^0L^{hcp_A3}_{Gd,Ru} = +25000 \times exp(-0.00013 \times T)$
		CM	${}^0L^{hcp_A3}_{Gd,Ru} = +25002.5 \times exp(-0.000323 \times T)$
Gd_3Ru < 1115	*$Gd_{0.750}$:$Ru_{0.250}$*	QM	${}^0G^{Gd_3Ru}_{Gd:Ru} - 0.750 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.750 \times {}^0G^{hcp_A3}_{Gd} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -21447.76 + 0.139 \times T$
		EM	${}^0G^{Gd_3Ru}_{Gd:Ru} - 0.750 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.750 \times {}^0G^{hcp_A3}_{Gd} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -15612 + 0.056 \times T$
		CM	${}^0G^{Gd_3Ru}_{Gd:Ru} - 0.750 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.750 \times {}^0G^{hcp_A3}_{Gd} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -15131.463 + 0.329 \times T$
$Gd_{73}Ru_{27}$ < 1150	*$Gd_{0.730}$:$Ru_{0.270}$*	QM	${}^0G^{Gd_{73}Ru_{27}}_{Gd:Ru} - 0.730 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.270 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.730 \times {}^0G^{hcp_A3}_{Gd} + 0.270 \times {}^0G^{hcp_A3}_{Ru} = -22787.436 + 0.0643 \times T$
		EM	${}^0G^{Gd_{73}Ru_{27}}_{Gd:Ru} - 0.730 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.270 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.730 \times {}^0G^{hcp_A3}_{Gd} + 0.270 \times {}^0G^{hcp_A3}_{Ru} = -16380.432 + 0.00703 \times T$
		CM	${}^0G^{Gd_{73}Ru_{27}}_{Gd:Ru} - 0.730 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.270 \times {}^{298.15}H^{hcp_A3}_{Ru}$
			$= +0.730 \times {}^0G^{hcp_A3}_{Gd} + 0.270 \times {}^0G^{hcp_A3}_{Ru} = -15626.807 + 0.0174 \times T$

Table 3 continued

Phase/temperature range (°C)	Thermodynamic model	Used model**	Thermodynamic parameters[a]
$Gd_2Ru < 1138$	$Gd_{0.667}$:$Ru_{0.333}$	QM	${}^0G^{Gd_2Ru}_{Gd:Ru} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.667 \times {}^0G^{hcp_A3}_{Gd} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -25661.855 + 0.0717 \times T$
		EM	${}^0G^{Gd_2Ru}_{Gd:Ru} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.667 \times {}^0G^{hcp_A3}_{Gd} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -18032.155 + 0.0821 \times T$
		CM	${}^0G^{Gd_2Ru}_{Gd:Ru} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.667 \times {}^0G^{hcp_A3}_{Gd} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -17284.12 + 0.0142 \times T$
$GdRu_2 > 1700$	$Gd_{0.333}$:$Ru_{0.667}$	QM	${}^0G^{GdRu_2}_{Gd:Ru} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.333 \times {}^0G^{hcp_A3}_{Gd} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -40312.55 + 0.0952 \times T$
		EM	${}^0G^{GdRu_2}_{Gd:Ru} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.333 \times {}^0G^{hcp_A3}_{Gd} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -26827.761 + 0.869 \times T$
		CM	${}^0G^{GdRu_2}_{Gd:Ru} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Gd} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru}$ $= +0.333 \times {}^0G^{hcp_A3}_{Gd} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -26110.798 + 0.414 \times T$

Units in J/mol-at and J/mol-at K. Reference states are solid (γCe) for *Ce*, solid (αNd) for *Nd*, and (*hcp_A3*) for *Dy, Gd,* and *Ru*, and (αSm) for *Sm* at *T*K.

Since the Neumann–Kopp Rule has been invoked, the formation properties are independent of temperature.

[a]Note that the enthalpies and entropies of formation of the intermetallic compounds are listed per mole of atoms.

*(V_a) for Vacancy.

**QM: Quadratic model (= Linear + $d*T^2$), EM: Exponential model, and CM: Combined model.

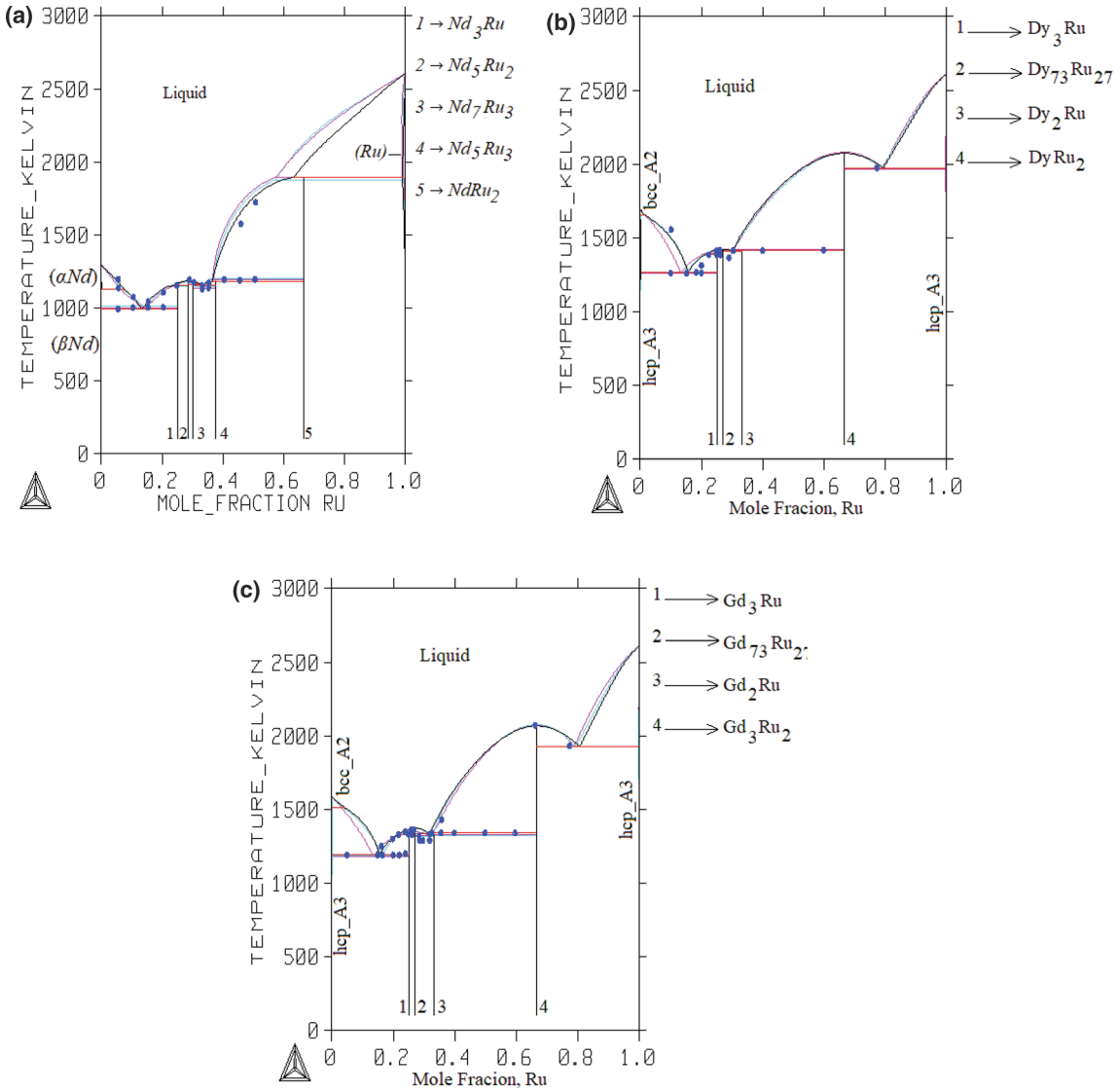


Fig. 3 (a) Superimposed *Nd-Ru* binary phase diagram calculated using our thermodynamic parameters in the three cases: black solid line: quadratic model, pink solid line exponential model, and light blue solid line: combined model, compared to (blue filled circle) experimental data established by Palenzona and Canepa.[23] (b) Superimposed *Dy-Ru* binary phase diagram calculated using our thermodynamic parameters in the three cases: black solid line: quadratic model, pink solid line exponential model, and light blue solid line: combined model, compared to (blue filled circle) experimental data established by Loebich and Raub.[26] (c) Superimposed *Gd-Ru* binary phase diagram calculated using our thermodynamic parameters in the three cases: black solid line: quadratic model, pink solid line: exponential model, and light blue solid line: combined model, compared to (blue filled circle) experimental data established by Loebich and Raub[26]

Palenzona[23] (see Fig. 1a). As we can see, the calculated liquidus in the *Nd*-rich part is in reasonable agreement with experimental data.[23] Then, within the three models, we have noted a better description concerning the calculated liquidus in the *Ru*-rich part. As shown in Table 4, the differences between the calculated and experimentally determined temperatures of the invariant reactions are within 6 K. A consistent thermodynamic assessment could not be obtained to agree with both temperature and composition for the $Liq \rightleftharpoons Nd_5Ru_3 + Nd_7Ru_3$ eutectic reaction calculated at (1155 K; 33.28 at.% *Ru*) with the quadratic model, and (1148 K; 33.96 at.% Ru) with a combined model instead of (1128 K; 34.00 at.% *Ru*) as reported by Ref 23. The temperature description for this reaction is much better using the exponential model (1134 K; 33.56 at.% *Ru*). In this study, it was found that the introduction of the further parameter *d* in ${}^0L$ expression can especially improve the description of the liquid phase. We have excluded the parameter *c* in this study because the *lnT* term leads to uncertainty when the temperature is extrapolated to 0 K. No solubility of *Nd* in solid *Ru* at the $NdRu_2 \rightleftharpoons hcp_A3 + Liq$ peritectic reaction at $\sim$ 1889 K

Table 4 Special points in the *Nd-Ru, Dy-Ru,* and *Gd-Ru* phase diagrams

Reaction	Composition, at.% *Ru*			Temperature, K	Reaction type	References
Nd-Ru binary system						
Liq ↔ *(βNd)*	0.00			1294	Melting	Ref 23, 25
						This work^{QM}
						This work^{EM}
						This work^{CM}
Liq + *(βNd)* ↔ *(αNd)*	07.00	0.00	0.00	1136	Catatectic	Ref 23, 25
	07.50			1127		This work^{QM}
	06.20			1128		This work^{EM}
	07.20			1128		This work^{CM}
Liq ↔ *(αNd)* + Nd_3Ru	13.00	0.00	25.00	998	Eutectic	Ref 23, 25
	13.37			992.8		This work^{QM}
	14.11			1000		This work^{EM}
	14.05			1008		This work^{CM}
Liq + Nd_5Ru_2 ↔ Nd_3Ru	24.00	28.60	25.00	1153	Peritectic	Ref 23, 25
	24.05			1155		This work^{QM}
	24.96			1152		This work^{EM}
	24.58			1149		This work^{CM}
Liq ↔ Nd_5Ru_2	28.60	28.60	…	1188	Congruent	Ref 23, 25
				1184		This work^{QM}
				1186		This work^{EM}
				1179		This work^{CM}
Liq + Nd_5Ru_2 ↔ Nd_7Ru_3	31.00	28.60	30.00	1163	Peritectic	Ref 23, 25
	32.52			1162		This work^{QM}
	31.90			1158		This work^{EM}
	31.84			1159		This work^{CM}
Liq ↔ $NdRu_x$ + Nd_7Ru_3	34.00	37.00	30.00	1128	Eutectic	Ref 23, 25
	33.28			1155		This work^{QM}
	33.56			1134		This work^{EM}
	33.96			1148		This work^{CM}
Liq + $NdRu_2$ ↔ $NdRu_x$	36.00	66.70	37.00	1193	Peritectic	Ref 23, 25
	36.64			1183		This work^{QM}
	36.73			1196		This work^{EM}
	36.98			1202		This work^{CM}
Liq + *(Ru)* ↔ $NdRu_2$	57.00	100	66.70	1889	Peritectic	Ref 23, 25
	63.24	99.06		1893		This work^{QM}
	57.69	99.49		1897		This work^{EM}
	57.50	99.46		1877		This work^{CM}
Dy-Ru binary system						
Liq ↔ *(βDy)*	0.00			1685.5	Melting	Ref 26
						This work^{QM}
						This work^{EM}
						This work^{CM}
(αDy) ↔ *Liq* + *(βDy)*	0.00	0.00	0.00	Does not appear	Catatectic	Ref 26
	1.56			1654		This work^{QM}
	1.34			1654		This work^{EM}
	1.33			1654		This work^{CM}

Table 4 continued

Reaction	Composition, at.% *Ru*			Temperature, K	Reaction type	References
Liq ↔ α(*Dy*) + Dy_3Ru	18.3	0.00	25.0	1263	Eutectic	Ref 26
	15.6			1263		This work^{QM}
	13.3			1260		This work^{EM}
	15.3			1263		This work^{CM}
Liq + $Dy_{73}Ru_{27}$ ↔ Dy_3Ru	22.3	27.0	25.0	1388	Peritectic	Ref 26
	21.7			1388		This work^{QM}
	21.3			1393		This work^{EM}
	22.2			1391		This work^{CM}
Liq ↔ $Dy_{73}Ru_{27}$	27.0	27.0	…	1423	Congruent	Ref 26
				1423		This work^{QM}
				1420		This work^{EM}
				1419		This work^{CM}
Liq ↔ $Dy_{73}Ru_{27}$ + Dy_2Ru	29.0	27.0	33.3	1363	Eutectic	Ref 26
	30.1			1377		This work^{QM}
	29.8			1406		This work^{EM}
	29.7			1389		This work^{CM}
Liq + $DyRu_2$ ↔ Dy_2Ru	30.3	66.7	33.3	1411	Peritectic	Ref 26
	30.3			1411		This work^{QM}
	30.6			1418		This work^{EM}
	30.3			1413		This work^{CM}
Liq ↔ $DyRu_2$	66.7	66.7	…	2073	Congruent	Ref 26
				2073		This work^{QM}
				2081		This work^{EM}
				2073		This work^{CM}
Liq ↔ (*Ru*) + $DyRu_2$	77.2	00.0	66.7	1973	Eutectic	Ref 26
	79.3			1973		This work^{QM}
	79.1			1962		This work^{EM}
	78.8			1973		This work^{CM}
Gd-Ru binary system						
Liq ↔ (β*Gd*)	00.0			1508.15	Melting	Ref 26
						This work^{QM}
						This work^{EM}
						This work^{CM}
(α*Gd*) ↔ *Liq* + (β*Gd*)	00.0	00.0	00.0	Does not appear	Catatectic	Ref 26
	03.84			1508		This work^{QM}
	03.07					This work^{EM}
	04.20					This work^{CM}
Liq ↔ α(*Gd*) + Gd_3Ru	14.8	00.0	25.0	1190	Eutectic	Ref 26
	15.6			1192		This work^{QM}
	13.5			1183		This work^{EM}
	16.3			1185		This work^{CM}
Liq + $Gd_{73}Ru_{27}$ ↔ Gd_3Ru	21.6	27.0	25.0	1328	Peritectic	Ref 26
	21.4			1327		This work^{QM}
	22.9			1331		This work^{EM}
	22.5			1332		This work^{CM}

Table 4 continued

Reaction	Composition, at.% *Ru*			Temperature, K	Reaction type	References
$Liq \leftrightarrow Gd_{73}Ru_{27}$	27.0	27.0	…	1373	Congruent	Ref 26
				1372		This work^{QM}
				1347		This work^{EM}
				1369		This work^{CM}
$Liq \leftrightarrow Gd_{73}Ru_{27} + Gd_2Ru$	29.0	27.0	33.3	1288	Eutectic	Ref 26
	31.6			1328		This work^{QM}
	31.9			1325		This work^{EM}
	31.9			1315		This work^{CM}
$Liq + GdRu_2 \leftrightarrow Gd_2Ru$	32.5	66.7	33.3	1338	Peritectic	Ref 26
	32.2			1340		This work^{QM}
	32.7			1327		This work^{EM}
	32.7			1321		This work^{CM}
$Liq \leftrightarrow GdRu_2$	66.7	66.7	…	2069	Congruent	Ref 26
				2066		This work^{QM}
				2072		This work^{EM}
				2075		This work^{CM}
$Liq \leftrightarrow (Ru) + GdRu_2$	77.2	0.00	66.7	1926	Eutectic	Ref 26
	80.6			1925		This work^{QM}
	78.8			1927		This work^{EM}
	79.2			1924		This work^{CM}
$Liq \leftrightarrow (Ru)$	100	100	…	2607	Melting	Ref 23, 25, 26, 59
						This work^{QM}
						This work^{EM}
						This work^{CM}

^{QM}: Quadratic model (= Linear model + $d*T^2$).

^{EM}: Exponential model.

^{CM}: Combined model.

has been reported. Still the *Ru*-rich side required more investigation, thus we have calculated a small solubility about 0.94, 0.51, and 0.54 at.% *Ru* at 1893, 1897, and 1877 K, respectively. The $(\alpha Nd) \rightleftharpoons (\beta Nd)$ allotropic transition was well reproduced at temperature 1136 K and 7 at.% *Ru*.

7.2 *Dy-Ru* Binary System

The superimposed calculated *Dy-Ru* phase diagram with the optimized thermodynamic parameters obtained using the three models is depicted in Fig. 3(b). The $Liq \rightleftharpoons Dy_{73}Ru_{27} + Dy_2Ru$ eutectic reaction was calculated at (1406 K; 29.8 at.% *Ru*) and (1389 K; 29.7 at.% *Ru*) versus (1363 K; 29 at.% *Ru*) according to Ref 26 in the case of exponential and combined model, respectively. A good agreement was achieved by quadratic model for this reaction with uncertainty of about 14 K. Except for this, we have observed for all three cases that all the invariant reactions in this system are well reproduced, with an uncertainty of about 6 K for temperature and 2 at.% *Ru* for composition. The $(\alpha Dy) \rightleftharpoons Liq + (\beta Dy)$ catatectic reaction was well reproduced at temperature of 1654 K and 1.56, 1.34, and 1.33 at.% *Ru* with quadratic, exponential, and combined models, respectively.

7.3 *Gd-Ru* Binary System

The superimposed calculated *Gd-Ru* phase diagram with the optimized thermodynamic parameters obtained using the three models is depicted in Fig. 3(c). A catatectic, two eutectic, and two peritectic reactions were calculated, as well as two congruent melting temperatures for $GdRu_2$ and $Gd_{73}Ru_{27}$. In the case of a similar system, it should be noted that the composition of the liquid phase in $Liq \rightleftharpoons Gd_{73}Ru_{27} + Gd_2Ru$ eutectic and $Liq + GdRu_2 \rightleftharpoons Gd_2Ru$ peritectic reactions is very close to that of the involved phases. Also, one can observe that the compositions of the three first compounds are close. Although, our optimization for this system leads to good agreement with the

experimental phase diagram shown in Fig. 1(c). Unfortunately, the $Liq \rightleftharpoons Gd_{73}Ru_{27} + Gd_2Ru$ eutectic reaction was not well reproduced.

7.4 Comparison of Our Thermodynamic Results and the Literature Data

All thermodynamic parameters were optimized together to give the best description of the considered system. In Table 2, the literature enthalpies of formation, measured by the calorimetric methods,[36,37,44,46,58,60] predicted by Ref 39, 40, 43, and calculated by the ab-initio method from the Materials Project,[41] OQMD,[42] Aflow[45] and Jarvis,[47] are compared with our optimized enthalpies of formation of the intermetallic compounds in the *Nd-Ru, Dy-Ru,* and *Gd-Ru* systems. Figure 4(a), (b), and (c) show the enthalpies, Gibbs energies, and relative Gibbs energies of formation of the intermetallic compounds existing in *Ln-Ru* systems referred to solid *Ru* and solid *Ln* that were optimized in this work by *CALPHAD* method with the three models. These thermodynamic values were compared with tabulated ones (Table 2). They are less negative than the predicted ones in Ref 39, 40, 43. The enthalpy of formation of the $LnRu_2$ is the highest. For that phase, the calculated enthalpy is in complete agreement with the experimental data.[36,37,44,46,58,60] All values are given in kJ/mol. The predicted standard enthalpies of formation of the five compounds in the *Nd-Ru* binary system were calculated using the Miedema model.[39,40,43] They are also shown. Large uncertainties were given to this model. The enthalpies of formation obtained from our ab-initio calculations are more exothermic than our calculated values from the three models. Other calculated values are obtained by ab-initio method using the Materials Project,[41] OQMD,[42] Aflow[45] and Jarvis.[47] Fig. 4(a), demonstrates the agreement of the value 17.2 ± 1.9 kJ/mol for Nd_5Ru_2 obtained by Ref 39 with our calculated value. For the $NdRu_2$ compound, it is evident that the measured values found by Guo and Kleppa[36] and Shilov et al.[37] are in agreement with our calculated values using the *CALPHAD* method. It should be noted that the data of Ref 36 and 37 apply to temperatures higher than 298.15 K (For $NdRu_2$, the value was obtained by *DTA* at 1030 K[37] and *DSC* at 1473 ± 2 K[36]). The difference between the data at 298.15 K and 1030 K or 1473 ± 2 K are usually not significant. So, the conclusions drawn from a comparison should still be valid.

In this section, it is appropriate to evaluate the thermodynamic stability of the Gibbs energy of $Ln_{1-x}Ru_x$ intermediate phases with respect to $Ln_{1-x'}Ru_{x'}$ and $Ln_{1-x''}Ru_{x''}$ neighboring phases. Such a comparison can be made based on the formula:[61]

$$\Delta_r G(x) = G(x) - (1-\eta) \times G(x') - \eta \times G(x'') \quad \text{(Eq 22)}$$

where $x'' \geq x \geq x'$ are the mole fractions of *Ru* in the corresponding phases and

$$\eta = (x - x')/(x'' - x').$$

The calculation results are shown in Fig. 4(a), (b), and (c). The calculation shows that the $LnRu_2$ phase is the most thermodynamically stable. According to the Landolt–Börnstein reference book,[62] all *Ln-Ru* phase diagrams, except for the *Eu-Ru* system, form the $LnRu_2$ phase. The heavy rare earth elements from *Gd* to *Lu* form a congruent melting phase $LnRu_2$ of the $MgCu_2$ type with a cubic cell *Fd3m*. All these phases melt approximately at temperatures of 2070 ± 50 K.

The phases based on $LnRu_2$ of the light elements from *La* to *Sm* have a hexagonal unit cell $P6_3/mmc$ of the $MgZn_2$ type. The nature of the melting of $LnRu_2$ phases of the $MgCu_2$ type remains open. According to Ref 62, the $LaRu_2$, $CeRu_2$, and $PrRu_2$ are formed by peritectic reaction in the temperature range 1400-1431, 1570-1573 K and 1954 K, respectively. According to Ref 23, the $NdRu_2$ phase has a peritectic melting point at *1635 K*, while according to Ref 63, it melts congruently at 2173 K. The $SmRu_2$ phase undoubtedly melts congruently at 2173 K.[59,64]

Based on the assessment of the thermodynamic stability of the $LnRu_2$ phases relative to the phases coexisting with it, and according to the formula from (22), the phases in the series $LaRu_2$, $CeRu_2$, $PrRu_2$, and $SmRu_2$ are the most thermodynamically stable and must have congruent melting temperatures, which increase almost linearly. At the same time, the Gibbs energies also change almost linearly depending on their melting points as shown in Fig. 5(a) and (b). Since the congruent temperatures of the $LnRu_2$ phases are unknown, and the coordinates of the $LnRu_2$-*Ru* eutectic are also unknown, we took these phases as incongruent. Such an assumption is possible if the congruent temperatures are close to the temperatures of peritectic reactions, and the eutectic are located near the $LnRu_2$ phases as in the case when *Ln* = *Nd*. Figure 5(c) shows tetrad-effect[65–67] for the $LnRu_2$ compounds vs N_{Ln}. The left part of the enthalpies of formation from *La* to *Gd* was described by following polynomial equation:

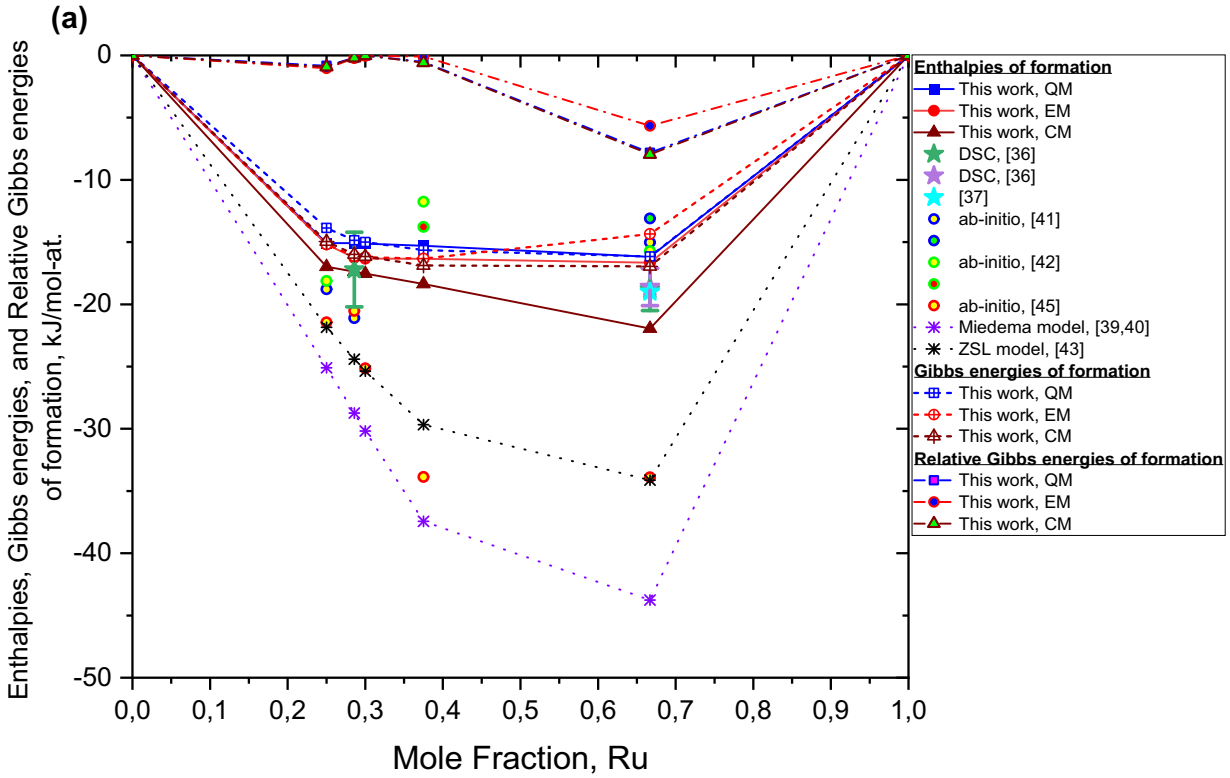
$$y = 0.080041 \times x^2 - 404.354 \times x + 404.354 \;\; ; \;\; 2 \times \sigma_0 = \pm\, 1.00 \text{ kJ/mol}.$$

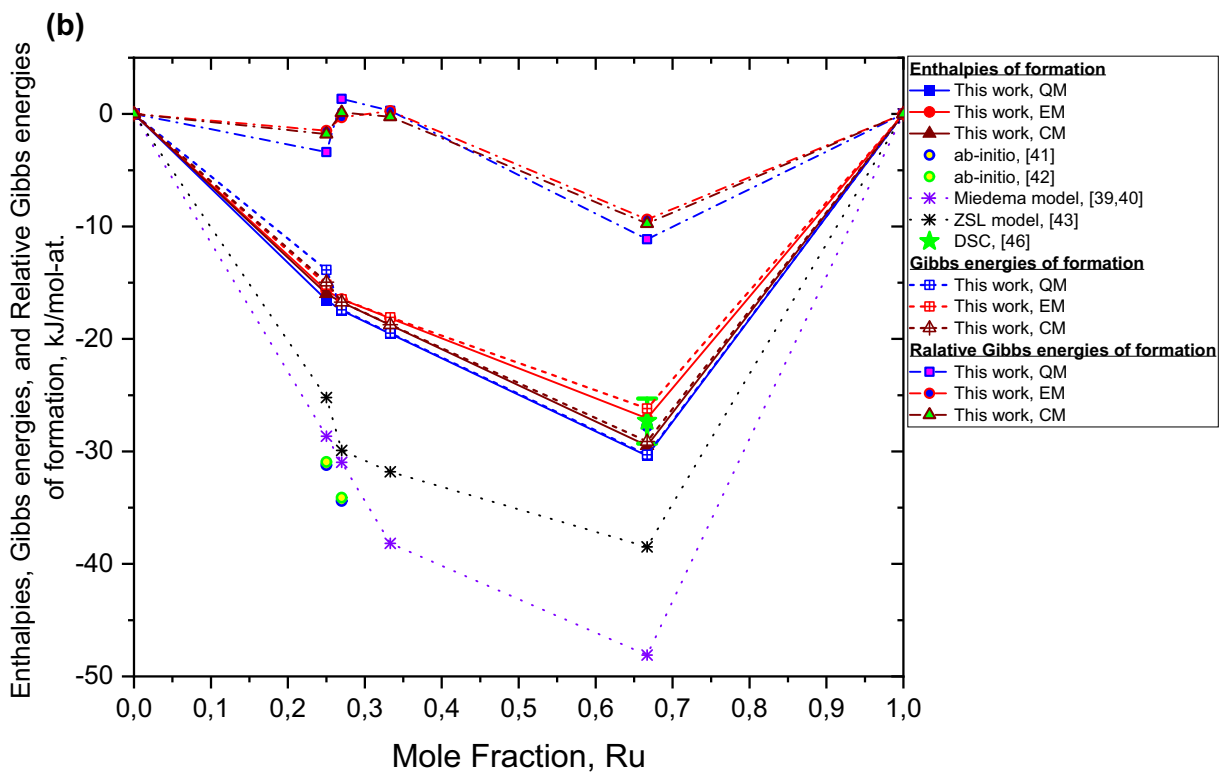
for the cubic crystalline structure ($MgCu_2$).

The right part of the enthalpies of formation from *Gd* to *Lu* was described by linear equation:

$$y = -0.1927 \times x - 13.9644 \; ; \; 2 \times \sigma_0 = \pm\, 1.50 \text{ kJ/mol}.$$

for the hexagonal crystalline structure ($MgZn_2$).


(a)
Enthalpies, Gibbs energies, and Relative Gibbs energies of formation, kJ/mol-at.
Mole Fraction, Ru
Enthalpies of formation
This work, QM
This work, EM
This work, CM
DSC, [36]
DSC, [36]
[37]
ab-initio, [41]
ab-initio, [42]
ab-initio, [45]
Miedema model, [39,40]
ZSL model, [43]
Gibbs energies of formation
This work, QM
This work, EM
This work, CM
Relative Gibbs energies of formation
This work, QM
This work, EM
This work, CM


(b)
Enthalpies, Gibbs energies, and Relative Gibbs energies of formation, kJ/mol-at.
Mole Fraction, Ru
Enthalpies of formation
This work, QM
This work, EM
This work, CM
ab-initio, [41]
ab-initio, [42]
Miedema model, [39,40]
ZSL model, [43]
DSC, [46]
Gibbs energies of formation
This work, QM
This work, EM
This work, CM
Ralative Gibbs energies of formation
This work, QM
This work, EM
This work, CM

◀**Fig. 4** (a) Relative Gibbs energies, Gibbs energies and standard enthalpies of formation for alloys of *Nd* with Ru compared with data for $NdRu_2$ and Nd_5Ru_2 by Guo and Kleppa[36] and by Shilov et al.,[37] our ab-initio calculations for the five intermetallic compounds together with ab-initio data from Ref 42, 45, 47, predicted values from the original model of Miedema and co-workers,[39,40] and Modified Miedema's (ZSL) model by Ref 43. (b) Relative Gibbs energies, Gibbs energies and standard enthalpies of formation for alloys of *Dy* with *Ru* compared with data for $DyRu_2$ by Guo and Kleppa,[46] with ab-initio data from Ref 42, 45, predicted values from the original model of Miedema and co-workers,[39,40] and Modified Miedema's (ZSL) model by Ref 43. (c) Relative Gibbs energies, Gibbs energies and standard enthalpies of formation for alloys of *Gd* with *Ru* compared with ab-initio data from Ref 47, 48, predicted values from the original model of Miedema and co-workers,[39,40] and Modified Miedema's (ZSL) model by Ref 43

plotted composition independent part of the excess Gibbs energy of mixing for the liquid phase as a function of the temperature from 0 to 20,000 K for L_0^{Φ} by applying the quadratic, exponential, and combined models, viz for *Nd-Ru* system, as follows:

- Quadratic model (= Linear model + $d_0^{\Phi} \times T^2$):

$$\Delta G^E_{x_{Ru}} = x_{Ru} \times (1 - x_{Ru}) \times (-1.17571E + 05 + 68.126 \times T - 0.0189 \times T^2) \quad \text{(Eq 23a)}$$

$$\Delta H = x_{Ru} \times (1 - x_{Ru}) \times (-1.17571\text{E} + 05 + 0.0189 \times T^2) \quad \text{(Eq 23b)}$$

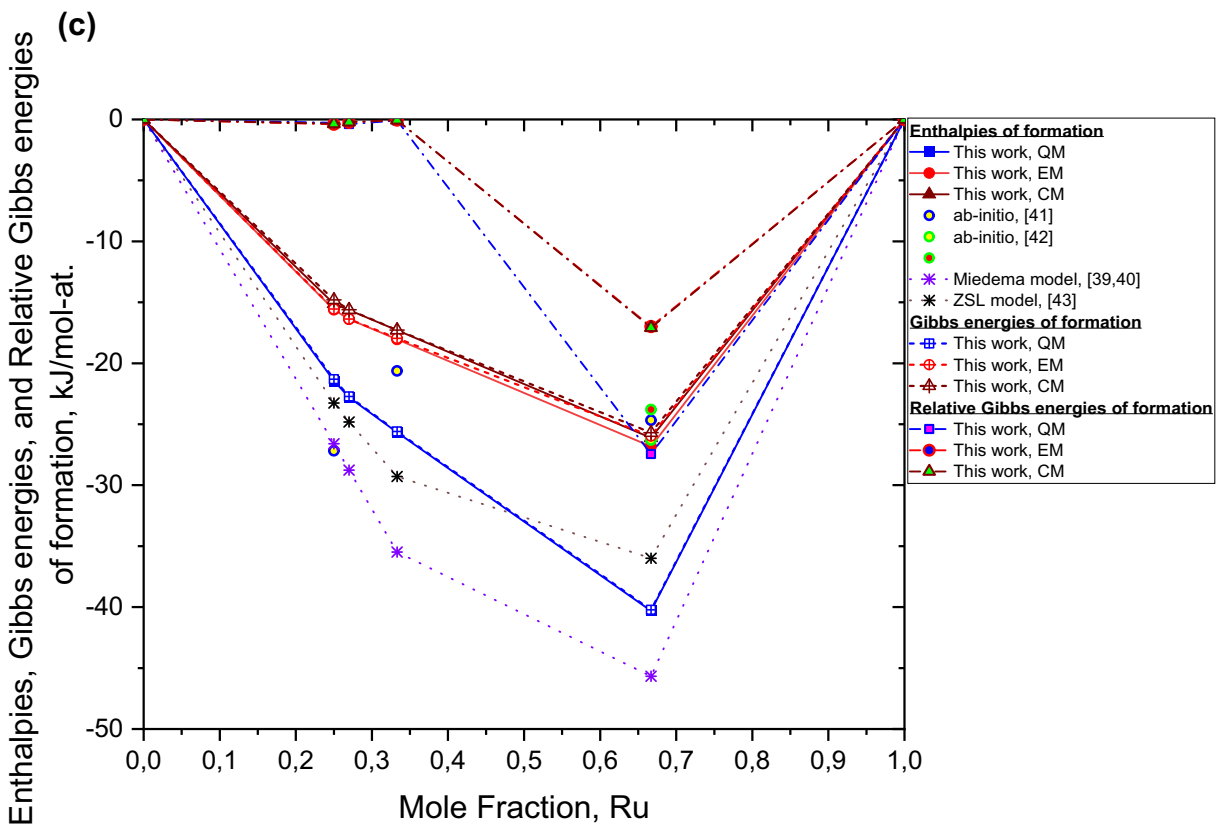


Fig. 4 continued

The $GdRu_2$ compound crystallizes in two forms: $MgCu_2$ and $MgZn_2$.

To ensure that our results fall in line with systematic tendency of other *Ln-Ru* studies, we have plotted the standard enthalpies of formation of the two common phases, $LnRu_2$ and Ln_3Ru, in *Ln-Ru* systems. The results are depicted in Fig. 6(a) and (b). Table 3 lists the final optimized thermodynamic parameters for the three systems, using the three models. Moreover, for all systems, we have

$$\Delta S^E_{x_{Ru}} = x_{Ru} \times (1 - x_{Ru}) \times (68.126 - 0.0378 \times T) \quad \text{(Eq 23c)}$$

$$\Delta C^E_{p,x_{Ru}} = -x_{Ru} \times (1 - x_{Ru}) \times (0.0378 \times T) \quad \text{(Eq 23d)}$$

- Kaptay model:

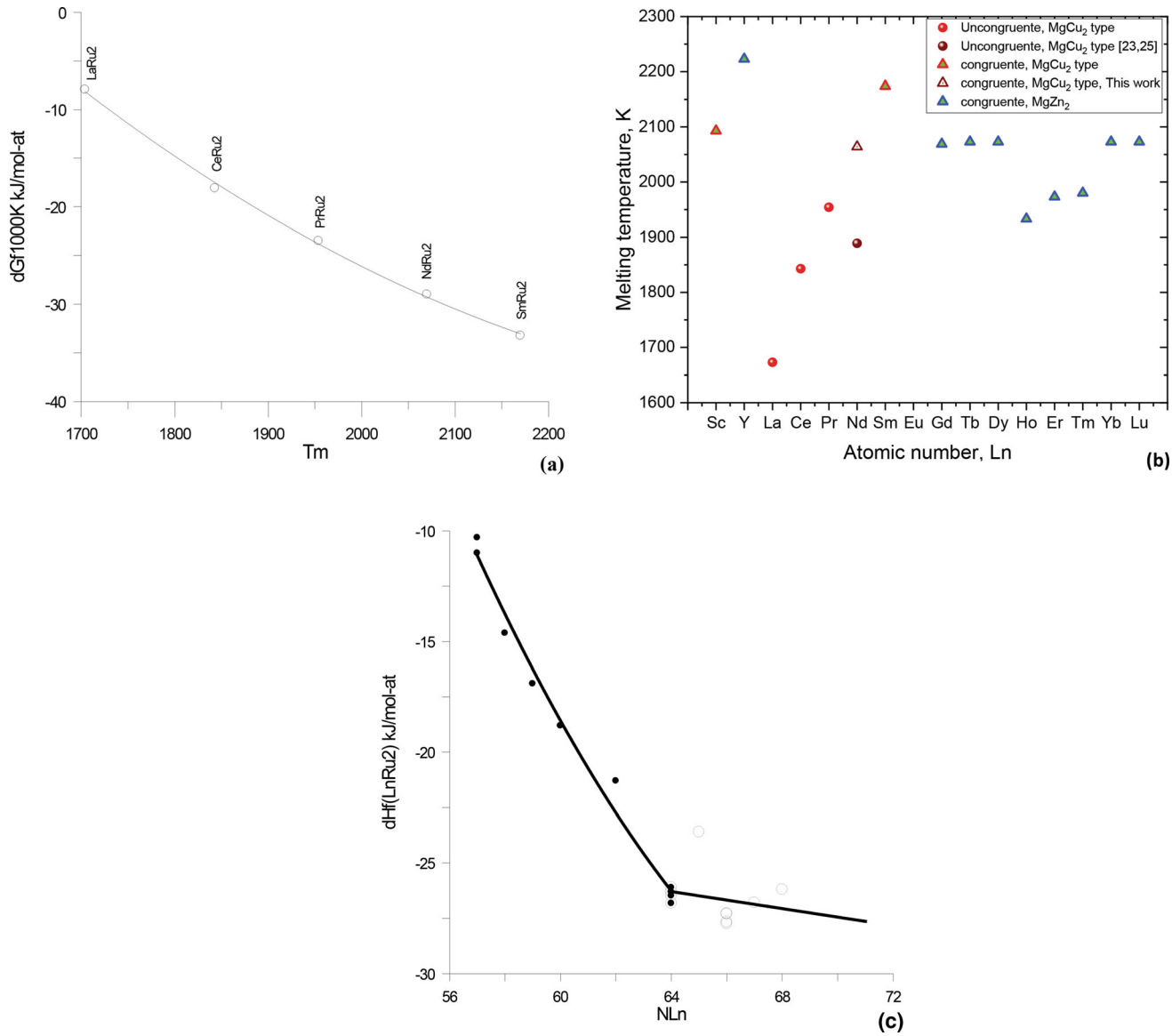


Fig. 5 The Gibbs energy vs. T_m (K) for $LaRu_2$, $CeRu_2$, $PrRu_2$, and $SmRu_2$ phases (a); T_m vs. atomic number (b). The Gibbs energy of the $PrRu_2$ and $NdRu_2$ phases were estimated. (c) Tetrad-effect[65–67] for the $LnRu_2$ compounds vs. N_{Ln}

$$\Delta G^E_{x_{Ru}} = x_{Ru} \times (1 - x_{Ru}) \times (-1.314\text{E} + 05 \times \text{EXP}(-6.53\text{E} - 04 \times T)) \quad \text{(Eq 24a)}$$

$$\Delta H = x_{Ru} \times (1 - x_{Ru}) \times (-1.314\text{E} + 05 \times (1 + 6.53\text{E} - 04 \times T) \times \text{EXP}(-6.53\text{E} - 04 \times T)) \quad \text{(Eq 24b)}$$

$$\Delta S^E_{x_{Ru}} = -x_{Ru} \times (1 - x_{Ru}) \times (85.83) \times \text{EXP}(-6.53\text{E} - 04 \times T) \quad \text{(Eq 24c)}$$

$$\Delta C^E_{p,x_{Ru}} = -x_{Ru} \times (1 - x_{Ru}) \times (56.04\text{E} - 03) \times T \times \text{EXP}(-6.53\text{E} - 04 \times T) \quad \text{(Eq 24d)}$$

- Combined model:

$$\Delta G^E_{x_{Ru}} = x_{Ru} \times (1 - x_{Ru}) \times (-1.41170\text{E} + 05 - 16.693 \times \text{T}) \times \text{EXP}(-7.66\text{E} - 04 \times T) \quad \text{(Eq 25a)}$$

$$\Delta H = x_{Ru} \times (1 - x_{Ru}) \times (-1.411\text{E} + 05 - 7.66\text{E} - 04 \times T(-1.411\text{E} + 05 - 16.693 \times \text{T}) \times \text{EXP}(-7.66\text{E} - 04 \times T)$$

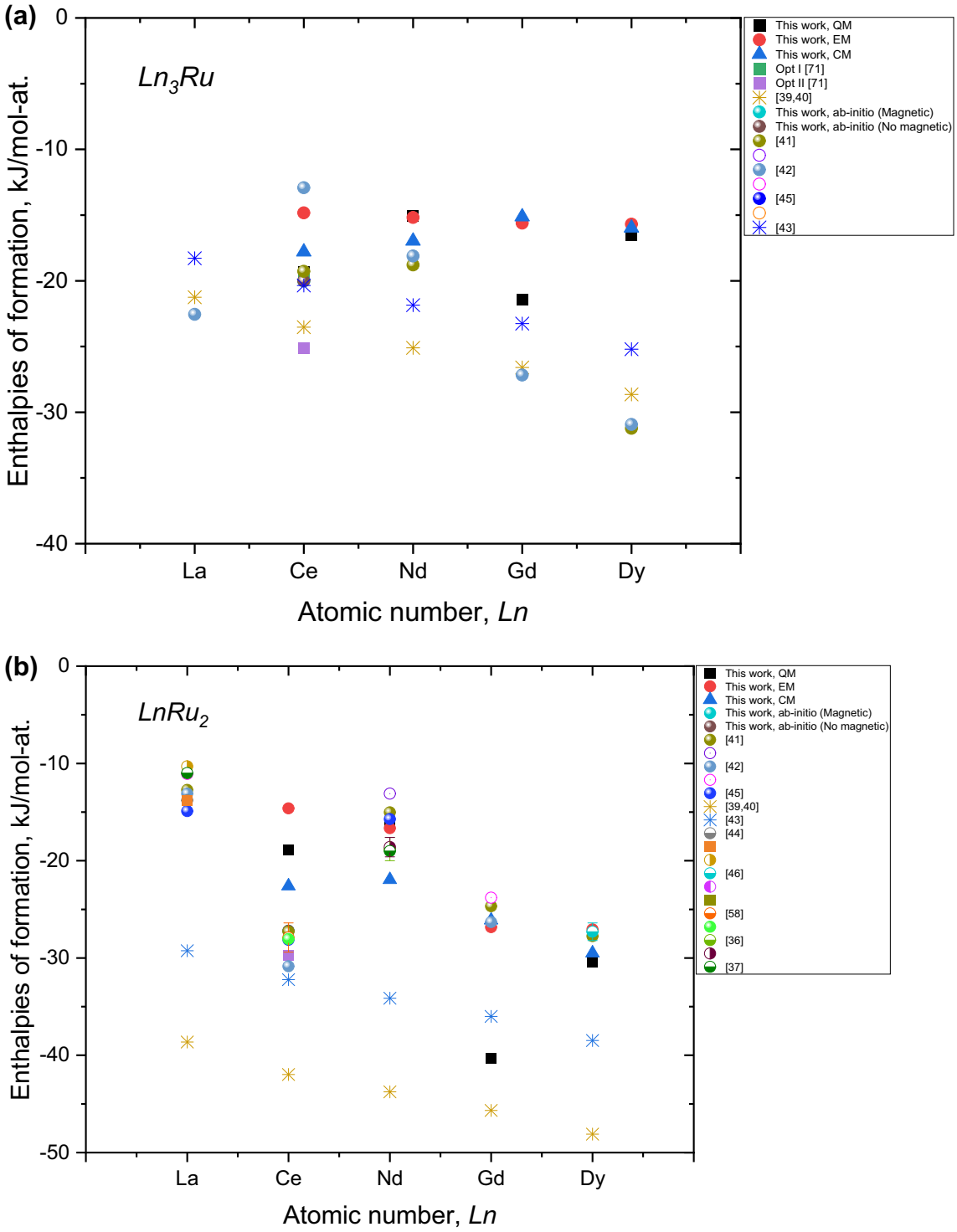


Fig. 6 (a) Optimised (*CALPHAD*) and calculated (ab-initio) standard enthalpies of formation of Ln_3Ru compared with calculated values by ab-initio from Ref 42, 45, 71 and with predicted ones from the Original Miedema model[39,40] and Modified Miedema's (ZSL) model by Ref 43. (b) Optimised (*CALPHAD*) and calculated (ab-initio) standard enthalpies of formation of $LnRu_2$ compared with calculated values by ab-initio from Ref 42, 45, 47 and with predicted ones from the Original Miedema model,[39,40] Modified Miedema's (ZSL) model by Ref 43, and experimental data from Ref 36–38, 44, 46, 58, 60

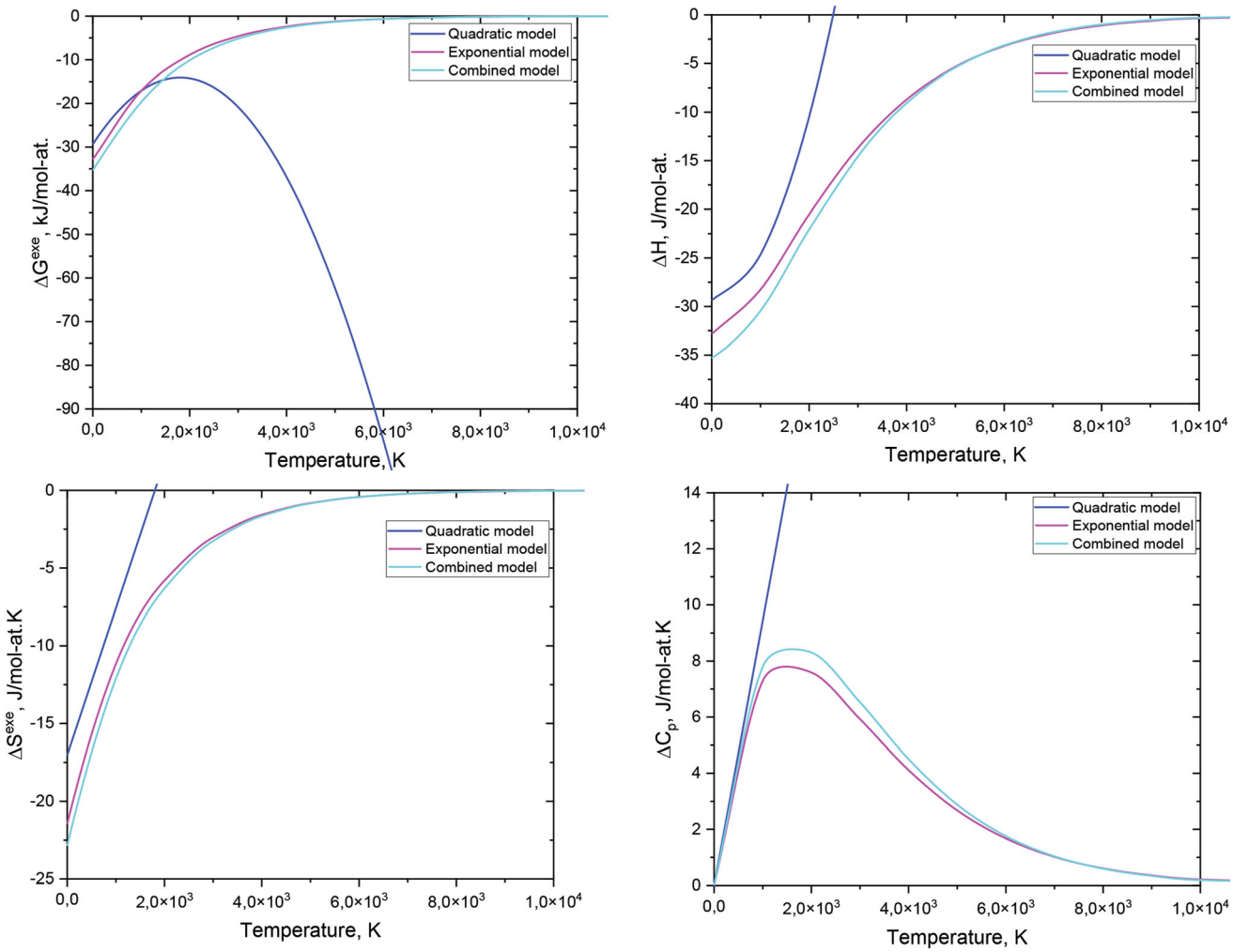


Fig. 7 Calculated *T*-dependences of the excess functions for an equiatomic solution for *neodymium-ruthenium* system of the quadratic model (Eq 14), of the exponential model (Eq 15), and of the combined model (Eq 16)

$$\Delta S^{E}_{x_{Ru}} = -x_{Ru} \times (1 - x_{Ru}) \times (-16.693 - 7.66\mathrm{E} - 04 \times (-1.411\mathrm{E} + 05 - 16.693 \times \mathrm{T})) \times \mathrm{EXP}(-7.66\mathrm{E} - 04 \times T)$$

$$\Delta C^{E}_{p,x_{Ru}} = -x_{Ru} \times (1 - x_{Ru}) \times (7.66\mathrm{E} - 04 \times T) \times (74.75 + 0.0127 \times T) \times \mathrm{EXP}(-7.66\mathrm{E} - 04 \times T) \quad \text{(Eq 25d)}$$

Figure 7, 8 and 9 demonstrate that in the three cases for all systems, the four excess properties are continuous functions of temperature in the whole temperature interval at any composition. At high temperature intervals, for exponential and combined models: in accordance with the rule of Lupis and Elliott[22] and later on reformulated in Ref 20, the excess Gibbs energy function tends toward zero. This behavior is in agreement with the requirements given in Ref 57. However for the quadratic model, this function changes its sign from negative to positive and becomes increasingly more positive, which is inconsistent.[20] Furthermore, the quadratic (with $d_0^{\Phi} = 0$) model predicts $\Delta C^{E}_{p,x_{Ru}} = 0$, which is in contradiction with the principal trend of the excess heat capacity of mixing as mentioned in Ref 21. At a medium temperature interval, all excess functions cross zero and go through maximum points. At a low temperature interval, the linear (quadratic with $d_0^{\Phi} = 0$) model provides a more negative excess Gibbs energy value compared to the exponential and combined models. This behavior changes at a intermediate temperature interval; the two latter models provide more negative excess Gibbs energy compared to the first one. The behavior observed with the quadratic model can be explained by using the temperature (T^2) factor d of Eq 2. As can be seen the ΔG^E and ΔC_p excess properties tend towards negatives values at intermediate temperatures,

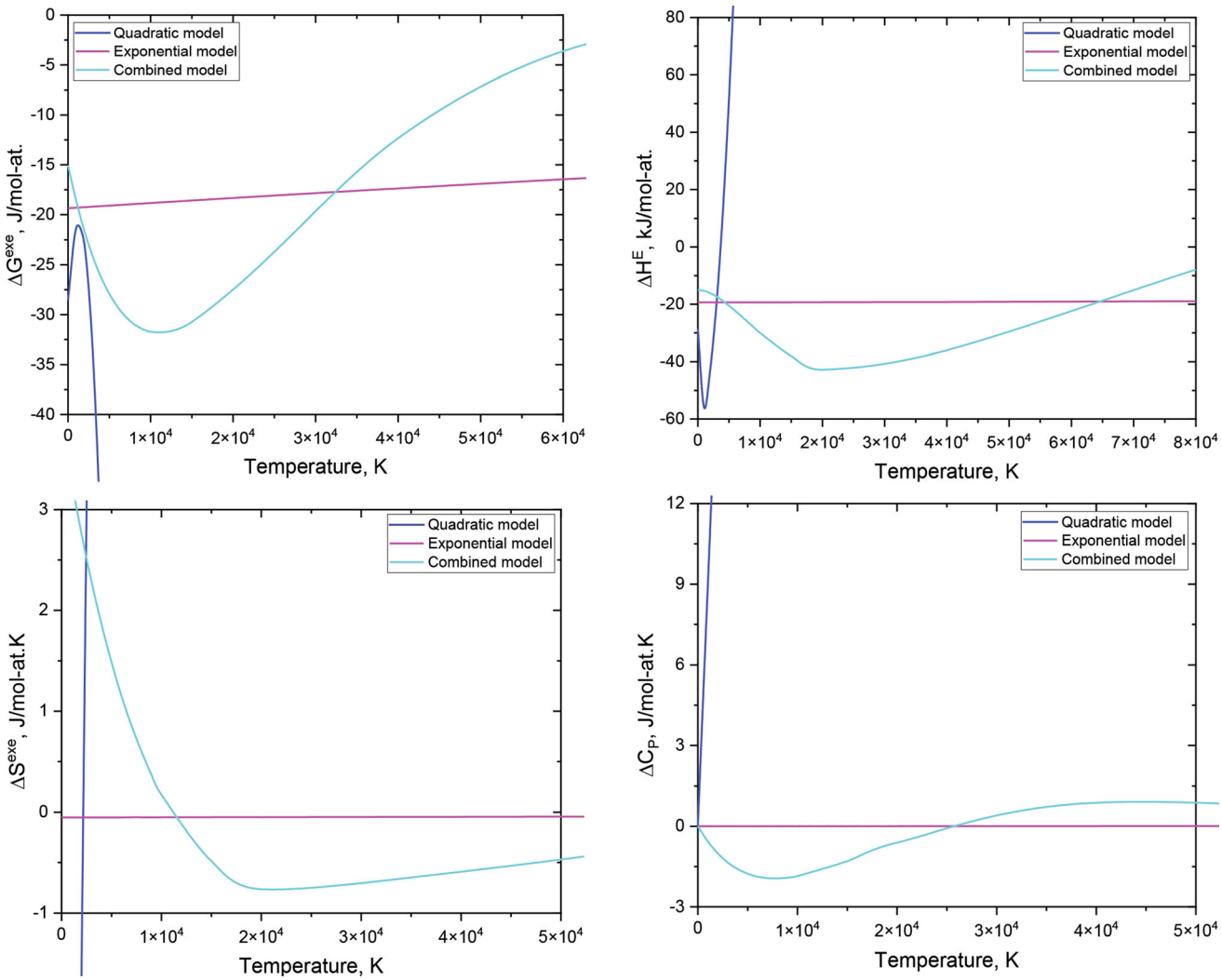


Fig. 8 Calculated *T*-dependences of the excess functions for an equiatomic solution for *dysprosium-ruthenium* system of the quadratic model (Eq 14), of the exponential model (Eq 15), and of the combined model (Eq 16)

while the ΔH function tends toward positives values when temperature increases. The same behavior was observed for ΔS^E function. Moreover, we have plotted the composition independent part of $d^2\Delta G/dx^2$ as a function of temperatures from 0 to 15,000 K for the three systems in the case of the three models. The results are shown in Fig. 10(a), (b), and (c). Although $d^2\Delta G/dx^2 > 0$ in the fully considered temperature range for the three systems, with the quadratic model, the occurrence of miscibility gap at high temperature ($a_0^\Phi < 0$ and $b_0^\Phi > 2 \times R$ are satisfied) was detected; this later formed at $T_{\min}$ (see Table 5). This is due to the optimized term (d) related to T^2 in the excess Gibbs energy of the liquid phase. The solution will not be stable at $T > a_0^\Phi/(2 \times R - b_0^\Phi)$. This finding is in accordance with Kaptay,[21] due to the application of an oversimplified quadratic model,[9–14,56] a considerable amount of systems from a high temperature miscibility gap. To systematically analyze the mixing enthalpy of liquid *Ln-Ru* (where $Ln = Ce$, Nd, Sm, Gd, and Dy) alloys, the mixing enthalpies of liquid *Y-Ru* and *La-Ru* alloys calculated by Selhaoui et al.[38,44,60] are also given in Fig. 11. The reference state is liquid for all elements at 1873 K and a pressure of 1e5 Pa. With the decrease of the *REE* atomic numbers, the calculated enthalpies of mixing of liquid in *Ln-Ru* ($Ln = Nd$, Gd, and Dy) alloys become much more negative. Besides, Takeuchi and Inoue[48] approximated the enthalpy of mixing with a formula of a subregular solution model with an interaction parameter Ω due to the

Table 5 The appearance temperature T_{min} of the artifact inverted miscibility gap due to the quadratic model for liquid phases in the considered *Ln-Ru* ($Ln = Nd,\ Gd,$ and Dy) systems

Quadratic model[9–14, 56]						
Ln-Ru system	a_0^{Φ} (J/mol-at)		b_0^{Φ} (J/mol-at K)		$T_{\min} = a_0^{\Phi}/(2 \times R - b_0^{\Phi})$ (K)	References
Nd-Ru	− 117,571.854	$a_0^{\Phi} < 0$	68.126	$b_0^{\Phi} > 2 \times R$	2283.070	Current work
Dy-Ru	− 114,253.113		46.898		3774.592	
Gd-Ru	− 164,983.881		74.404		2217.406	

where R = 8.31447 J/mol K is the universal gas constant.

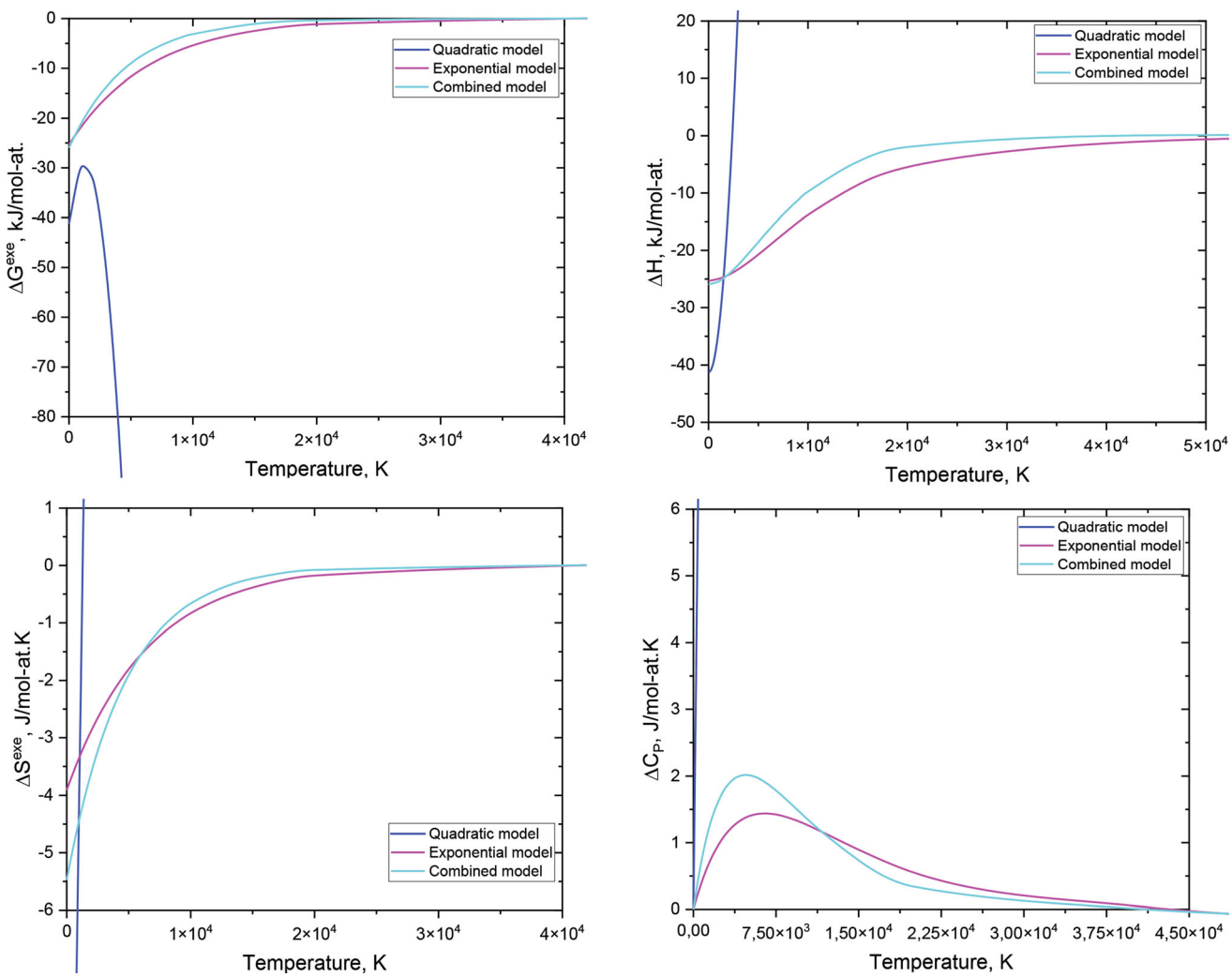


Fig. 9 Calculated *T*-dependences of the excess functions for an equiatomic solution for *gadolynium-ruthenium* system of the quadratic model (Eq 14), of the exponential model (Eq 15), and of the combined model (Eq 16)

awkward procedures for evaluating surface fraction term in Miedema's scheme. Thus, we have used this approximation to calculate the enthalpy of mixing of the liquid phase in our system (where Ω_k with k = 0,1,2, and 3 values were taken from Ref 48) together with one predicted by the Miedema model.[39,40,43] It can be seen that the calculated

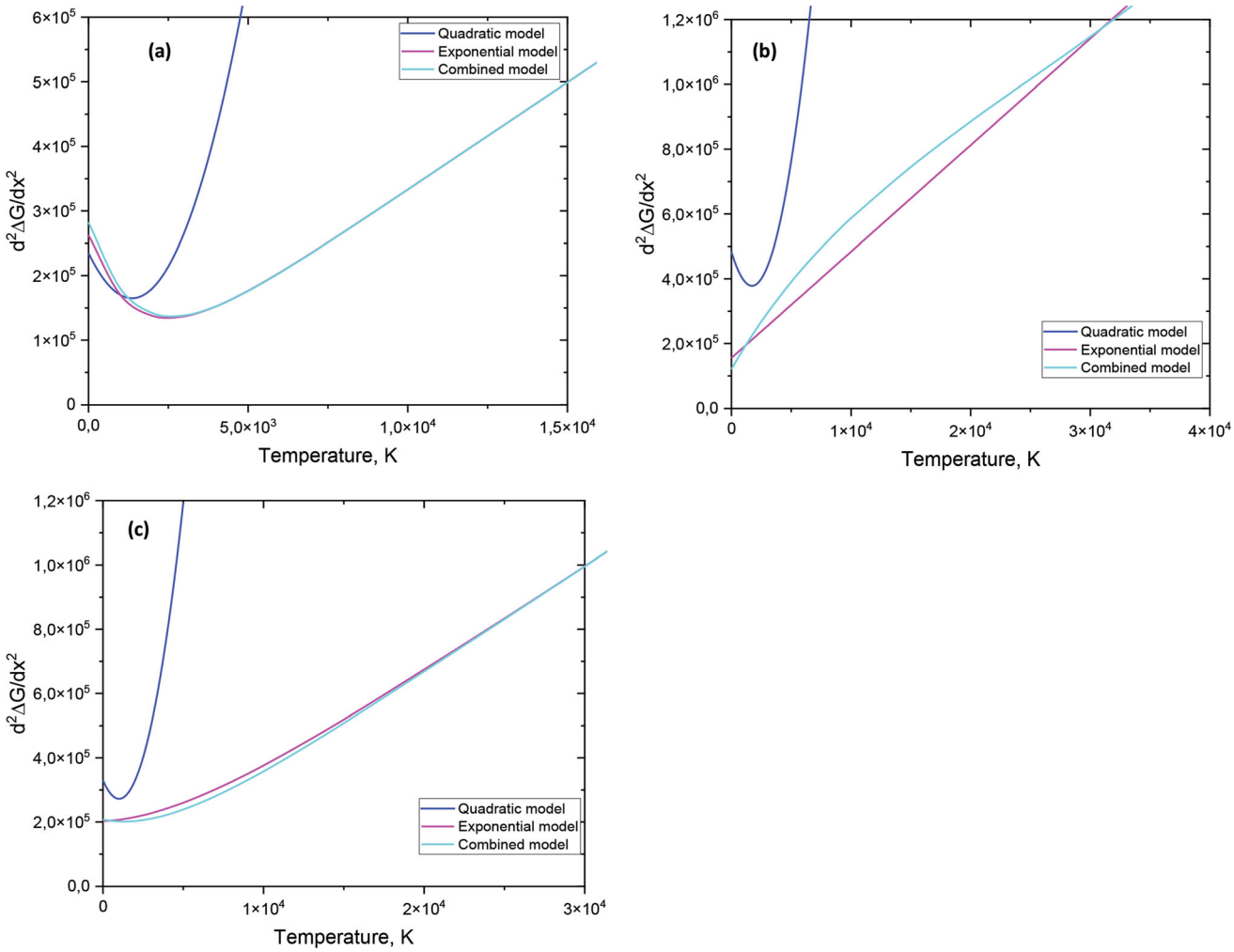


Fig. 10 Calculated second derivative vs. temperature for an equiatomic solution for *neodymium-ruthenium* (a), *dysprosium-ruthenium* (b), and *gadolinium-ruthenium* (c) systems of the exponential model (pink solid line), combined model (light blue solid line), and quadratic model (dark blue solid line)

enthalpies of mixing of liquid *Nd-Ru* alloys have a negative deviation, and show minimum values in the composition range of 40-50 at.% *Ru*. From Fig. 12, one can observe that the light rare earths *Sc*,[68,69] *Y*,[60] *La*,[44] *Nd*[23,25] show a similar behavior both in the shape of the phase diagrams and in the formation of the intermediate phases. The *Ln-Ru* systems, from 0 to 40 at. % *Ru* are characterized by relatively low melting points with the formation of a large number of intermediate phases. The melting points then rise sharply to the melting point of ruthenium with the formation of the peritectic reaction of the Laves phase $LnRu_2$: light rare earths with the *C15* ($MgCu_2$-type structure) and heavy rare earths with the *C14* ($MgZn_2$-type structure). This situation, however, might change for *Gd* and *Dy*.[26]

7.5 On the Abilities and Limitations of the Quadratic (= Linear + $d_0^{\Phi} \times T^2$), Exponential, and Combined Models

For most thermodynamic systems, published in literature, the quadratic (with $d_0^{\Phi} = 0$) model is frequently used and offers a reasonable approximation in a narrow temperature range around the liquidus. Nevertheless, it predicts that at infinitely high temperatures, the interaction energies and excess Gibbs energy of mixing will become infinitely larger rather than reaching zero. Hence, when extrapolated to high temperatures, this model produces unreasonable results, viz artificial inverted miscibility gaps. The exponential model was developed by Kaptay[21] to overcome this issue. Nonetheless, according to Schmid-Fetzer et al.,[13] it could lead to low temperature artifacts. Furthermore, this model is not able to reproduce data measured for some systems in which the excess Gibbs energy

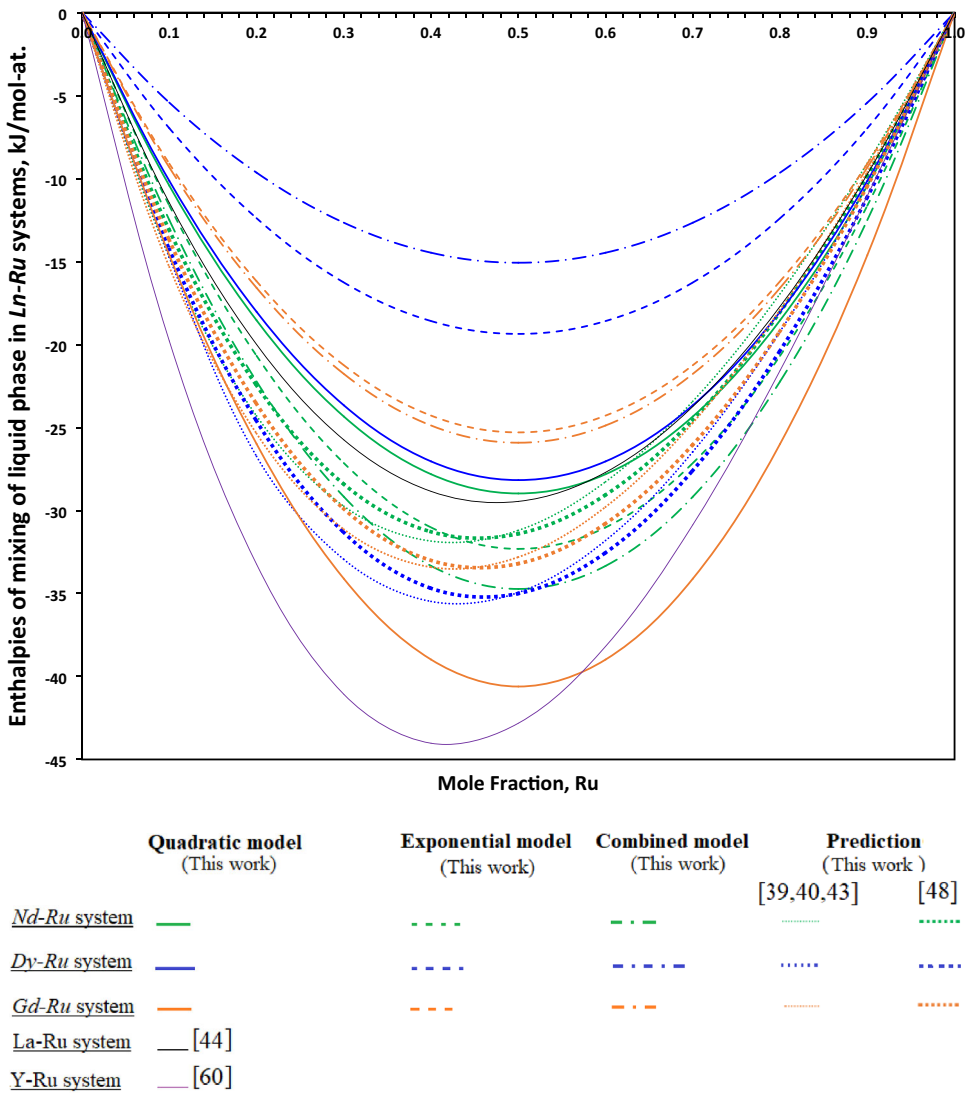


Fig. 11 Calculated liquid enthalpies of mixing at $T = 1000$ K in the binary systems *Ln-Ru*, where *Ln* = *Y*,[60] *La*,[44] *Nd*,[23,25] *Dy*,[26] and *Gd*[26] compared with those predicted by Miedema model[39,40,43] and approximated by Takeuchi and Inoue,[48] The reference states are: liquid for all *Ln* and *Ru*

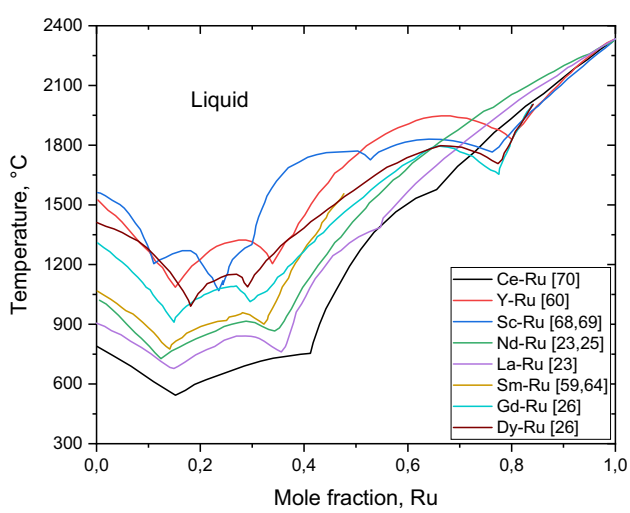


Fig. 12 Liquidus curves of the *Ln-Ru* binary systems, with *Ln* = *Sc*,[68,69] *Y*,[60] *La*,[23] *Ce*,[70] *Nd*,[23,25] *Sm*,[59,64] *Gd*,[26] and *Dy*[26]

of mixing is different from the sign of both the heat of mixing and the excess entropy of mixing. Compared to the exponential model, the combined model is more efficient since it can accurately reproduce experimental data of any real system, even these exceptional systems. There will not be any low-*T* or high-*T* artifacts generated by this model. In the current work and regarding the scarcity of the available data in literature especially for the liquid phase, precisely three first models have been the main aim of our analysis and attention.

8 Conclusion

For the three systems, the quadratic model for the temperature dependence of the interaction energies applied to liquid alloys leads to an unwanted inverted miscibility gap at high temperature or at intermediate temperature. We

have avoided this problem by replacing this model with an exponential model. In this study, to avoid *CALPHAD* artifacts at any temperature, we have demonstrated that the combined quadratic (with $d_0^{\Phi} = 0$)-exponential model is a proven tool. Although there is a limited amount of experimental information such as in the enthalpies of mixing of the liquid phases, self-consistent thermodynamic descriptions were obtained; these were used to calculate the corresponding phase diagrams, which are generally in reasonable agreement with available literature data. To verify the predictions made in the present study and also clarify some known discrepancies for these binaries, we suggest as a future experiment, the investigation of the *Ru*-rich side.

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